

WHAT MATTERS FOR REPRESENTATION ALIGNMENT: GLOBAL INFORMATION OR SPATIAL STRUCTURE?

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ABSTRACT

Representation alignment (REPA) is among the most effective techniques for accelerating generative training, distilling the representations of a strong pretrained vision encoder into the intermediate features of a diffusion transformer. Yet the mechanism underlying its gain has remained unresolved. We settle a fundamental question about that mechanism: which property of the target representation actually governs generation—its *global* semantic content (e.g., as measured by ImageNet-1K accuracy) or its spatial structure (i.e., the pairwise cosine similarity among patch tokens)? The prevailing assumption holds that stronger global semantic performance yields a better alignment target. Through a large-scale empirical analysis spanning 27 vision encoders and three diffusion model scales, we overturn this assumption and establish the opposite: it is spatial structure, not global performance, that determines the generative utility of a target representation. To render this dependence quantitative, we introduce a family of fast-to-compute spatial self-similarity metrics whose values correlate far more strongly with generation quality (Pearson $|r| > 0.85$) than does linear probing accuracy ($|r| = 0.26$). Building directly on this account, we derive two minimal modifications (<4 lines of code) that explicitly accentuate the transfer of *spatial* information: we replace the standard MLP projection layer in REPA with a single convolutional layer that preserves neighborhood relationships by construction, and we apply a spatial normalization layer to the external representation to sharpen its spatial contrast. The resulting method, iREPA, consistently accelerates the convergence of REPA across a diverse range of vision encoders, diffusion model sizes, and training recipes—spanning REPA, REPA-E, MeanFlow, and JiT. Together, these results reframe the working mechanism of representation alignment and chart a concrete path toward exploiting it for markedly more efficient training of generative models.

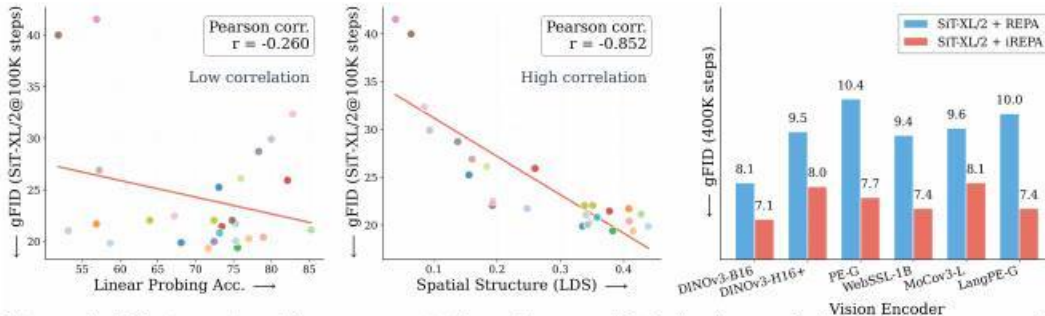


Figure 1: **What matters for representation alignment?** *Left:* A correlation analysis across 27 diverse vision encoders. Contrary to the prevailing assumption, it is the spatial structure of a target representation—and not its global performance, as measured by linear probing accuracy—that governs generation performance. *Right:* Building directly on this finding, we introduce two minimal modifications that accentuate the transfer of spatial features from the target representation to the diffusion model; the resulting method, iREPA, consistently accelerates the convergence of REPA across diverse settings.



Figure 2: **Motivating examples — spatial structure matters.** These comparisons expose a clear inversion between ImageNet accuracy and generation quality. *Left:* PE-G, despite far higher validation accuracy (82.8% vs. 53.1%), yields worse generation than SpatialPE-B (Bolya et al., 2025). *Right:* WebSSL-1B (Fan et al., 2025) likewise attains much higher global performance (76.0% vs. 53.1%) yet generates worse. **Spatial Self-Similarity:** Spatial structure, by contrast, is the reliable predictor of generation quality; see §3 for the spatial structure metric. All results reported at 100K iterations using SiT-XL/2 and REPA.

1 INTRODUCTION

Representation alignment has become one of the most effective techniques for accelerating the training of diffusion transformers (Ma et al., 2024; Peebles & Xie, 2023). By aligning the internal representations of a diffusion transformer with those of a pretrained self-supervised visual encoder, recent methods deliver substantial gains in both convergence speed and final generation quality (Yu et al., 2024; Leng et al., 2025a). These empirical successes are, however, only partially understood: the precise mechanism through which self-supervised features enhance diffusion training has remained an open problem. That gap poses a fundamental question—does the improvement arise principally from richer global semantic information, as commonly quantified through linear probing accuracy, or from the faithful transfer of spatial structure, characterized by the pairwise relationships among patch token representations? Resolving this question is what our work sets out to do.

The answer carries direct practical consequence, since it determines how practitioners should select a target representation and how they should extract its full benefit. The prevailing assumption holds that an encoder’s suitability for representation alignment tracks its ImageNet-1K validation accuracy—a proxy for global semantic understanding (Oquab et al., 2024; Chen et al., 2021). Under this view, target representations with higher ImageNet accuracy are expected to yield better generation (Yu et al., 2024). The assumption is self-reinforcing: gains in the linear probing accuracy of diffusion features are routinely invoked as evidence for the effectiveness of representation alignment, further entrenching global semantic information as the presumed primary driver of improvement. The following passage from REPA (Yu et al., 2024) crystallizes this conventional understanding:

“When a diffusion transformer is aligned with a pretrained encoder that offers more semantically meaningful representations (i.e., better linear probing results), the model not only captures better semantics but also exhibits enhanced generation performance, as reflected by improved validation accuracy with linear probing and lower FID scores.”

In this paper, we overturn this conventional wisdom through a systematic investigation of what truly drives representation alignment: global semantic information or spatial structure. Grounding the analysis in a large-scale empirical study across a broad and diverse set of vision encoders—including recent large-scale vision foundation models such as WebSSL (Fan et al., 2025), DINOv3 (Siméoni et al., 2025), perception encoders (Bolya et al., 2025), and C-RADIO (Heinrich et al., 2024)—we arrive at three findings that jointly reshape this understanding.

Higher validation accuracy does not imply a better representation for generation. Contrary to the prevailing assumption, vision encoders with stronger global semantic performance—as measured by ImageNet-1K linear probing accuracy—frequently *underperform* as alignment targets. Consider PE-Spatial-B, a spatially tuned model derived from PE-Core-G (Bolya et al., 2025). Despite attaining far lower patch-token validation accuracy¹ than PE-Core-G (53.1% vs. 82.8%), PE-Spatial-B yields markedly better generation under REPA (Figure 2). WebSSL-1B (Fan et al., 2025) exhibits precisely the same inversion: substantially higher global accuracy (76.0% vs. 53.1%), yet worse generation. As

¹Following REPA (Yu et al., 2024), we measure the global semantic performance of an external representation via linear probing accuracy on patch tokens, since only patch tokens are used for representation alignment.

we demonstrate in §2, this inversion is not anecdotal but recurs consistently across a wide range of target representations, families, and controlled interventions—signaling a governing principle behind how representation alignment benefits diffusion training.

Spatial structure, not global information, determines generation performance. To render this insight quantitative, we introduce a family of straightforward, fast-to-compute metrics that measure the spatial self-similarity structure (Shechtman & Irani, 2007) among patch tokens (§3). Using these metrics, we conduct a large-scale correlation analysis relating REPA generation FID to linear probing accuracy and to spatial structure across 27 vision encoders and three diffusion model scales (SiT-B, SiT-L, SiT-XL). The result is decisive: every spatial structure metric correlates strongly with generation FID (Pearson $|r| > 0.852$), far surpassing the near-negligible predictive power of ImageNet-1K validation accuracy ($|r| = 0.26$). Complementary controlled experiments reinforce the conclusion, establishing that representations carrying very little global information can nonetheless deliver substantial REPA gains. SAM2, for instance, yields better generation than encoders with roughly 60% higher ImageNet accuracy (Fig. 3); and even classical, purely spatial descriptors such as SIFT (Lowe, 1999) and HOG (Dalal & Triggs, 2005)—though not the strongest targets—produce meaningful gains under representation alignment (§3).

Accentuating spatial features improves convergence. Motivated by this mechanistic account, we derive two minimal modifications (<4 lines of code) that explicitly amplify the transfer of spatial information from the target representation to the diffusion model. First, we introduce a spatial regularization layer that sharpens the spatial contrast of the target representations. Second, we identify the standard MLP projection layer—used to map diffusion features onto the target representation dimensionality—as a systematic source of spatial information loss (Figure 6a), and replace it with a single convolutional layer that preserves local spatial relationships by construction. The effect is both pronounced and consistent: the resulting method, iREPA, reliably accelerates the convergence of REPA across diverse encoders, diffusion model sizes, and training recipes, including REPA-E (Leng et al., 2025a) and MeanFlow (Geng et al., 2025) with REPA.

The principal contributions of this paper are as follows:

- **A mechanistic account of representation alignment.** Through a large-scale empirical analysis spanning 27 vision encoders and three diffusion model scales, we establish that spatial structure—not global semantic information—is the property that drives the effectiveness of representation alignment, overturning the prevailing accuracy-centric view.
- **A predictive spatial structure metric.** We introduce a straightforward, fast-to-compute Spatial Structure Metric (SSM) that correlates substantially more strongly with downstream FID than linear probing accuracy (Pearson $|r| > 0.85$ vs. 0.26), and that furnishes a principled explanation for generation trends left unexplained by prior work.
- **iREPA, a minimal and general recipe.** We translate this mechanistic insight into two simple modifications that accentuate the transfer of spatial information from the target representation to the diffusion features. The resulting method delivers consistent convergence-speed improvements over REPA across variations in target representation, diffusion model architecture, and training recipe (REPA, REPA-E, MeanFlow with REPA, and JiT with REPA), demonstrating that the principle is recipe-agnostic rather than specific to any single setup.

2 MOTIVATION: GLOBAL INFORMATION MATTERS LESS

We begin with a series of motivating observations that converge on a single conclusion: the global performance of a target representation (ImageNet-1K accuracy) does not translate into better generation with REPA. Each observation resists explanation under the conventional, accuracy-centric view; in §3 we show that all of them are precisely accounted for once the spatial structure of the target representation is measured.

Recent vision encoders. We first examine a range of recent vision encoders, including Perception Encoders (Bolya et al., 2025), WebSSL (Fan et al., 2025), and DINOv3 (Siméoni et al., 2025). Consider PE-Spatial-B (80M), a compact spatially tuned model derived from PE-Core-G (1.88B) (Bolya et al., 2025). As Figure 2 shows, PE-Core-G attains far higher ImageNet-1K accuracy (82.8% vs. 53.1%) yet performs distinctly worse as the REPA target representation (FID 32.3 vs. 21.0)—a substantial margin, despite the more than 20-point accuracy advantage. WebSSL-1B (1.2B) exhibits

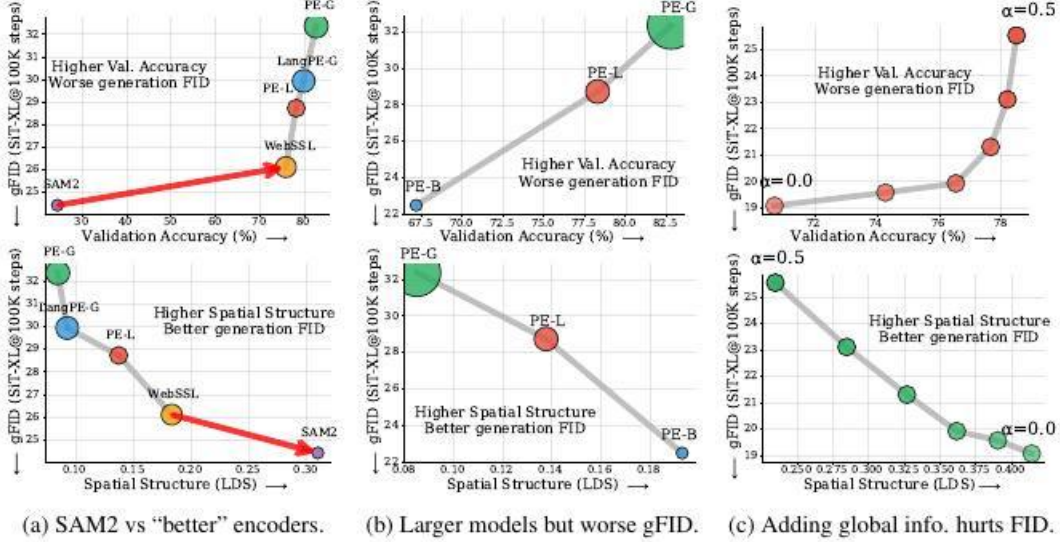


Figure 3: **Higher global information does not imply better REPA performance.** **Top row:** Across several distinct settings, global performance fails to track generation FID under REPA. (a) SAM2-S, with a validation accuracy of only 24.7%, achieves better generation with REPA than models with $\sim 60\%$ higher validation accuracy. (b) Larger encoders within the same family can attain higher validation accuracy yet worse generation. (c) Injecting global information into patch tokens via the CLS token raises global performance but degrades generation. **Bottom row:** Spatial structure, and not global performance, is the reliable indicator of generation quality; see §3 for the large-scale analysis across spatial structure metrics. All results are reported w/o classifier free guidance, SiT-XL/2 w/ REPA and 250 NFE (Yu et al., 2024) for inference.

the identical inversion: despite substantially higher ImageNet-1K accuracy (76.0% vs. 53.1%), it yields worse generation as the REPA target representation (FID 26.1 vs. 21.0). In both cases the more accurate encoder is the weaker alignment target, directly contradicting the accuracy-centric assumption.

SAM2 outperforms vision encoders with far higher ImageNet-1K accuracy. To probe how little global information a target representation actually requires, we analyze the SAM2-S vision encoder (46M) (Ravi et al., 2024)—a small model carrying very little global information, with a validation accuracy of only 24.1% (Fig. 3a). Even so, when used for REPA, SAM2-S achieves a better FID than vision encoders with dramatically higher ImageNet-1K accuracy, PE-Core-G (82.8%) among them. That a target with roughly 60 points less accuracy can nonetheless surpass far stronger encoders indicates that global performance is, at best, an unreliable determinant of REPA effectiveness.

Larger models within the same encoder family can yield similar or worse generation. It is commonly assumed that larger models within an encoder family carry better representations, as reflected in ImageNet-1K accuracy. For representation alignment, however, the larger variants often produce comparable (DINOv2) or even worse (PE, C-RADIO) generation performance (Fig. 3b)—the opposite of what the accuracy-centric view predicts. (Yu et al., 2024) report a similar observation for DINOv2, attributing it to the conjecture that “all DINOv2 models being distilled from the DINOv2-g model and thus sharing similar representations.” We later show that spatial structure furnishes a more principled and general explanation of this trend, one that also covers the PE and C-RADIO families (§3).

Injecting more global information can hurt generation. To isolate the causal effect of global information, we conduct a controlled experiment that deliberately injects global semantics—via the CLS token—into the local patch tokens (using DINOv2 as the encoder). The CLS-token mixing operation is $\mathbf{p}_i^{\text{new}} = \mathbf{p}_i + \alpha \cdot \mathbf{c}$, where $\mathbf{p}_i$ denotes the i^{th} patch token, $\mathbf{c}$ the CLS token, and $\alpha \in [0, 0.5]$ modulates the mixing strength. As Figure 3c shows, increasing α from 0 to 0.5 raises linear probing accuracy monotonically from 70.7% to 78.5%, while generation quality degrades in lockstep, FID worsening from 19.2 at $\alpha = 0$ to 25.4 at $\alpha = 0.5$. Adding global information therefore improves the very proxy presumed to predict REPA quality while actively harming the outcome that proxy is meant to predict—a monotonic divergence that isolates global information as a cause of degradation rather than improvement.

Taken together, these observations establish that the global performance of a representation is not a reliable indicator of its REPA effectiveness. We show next (§3) that spatial structure supplies exactly the predictive signal that global performance lacks.

- **Finding 1.** *Stronger global semantic information (e.g., ImageNet-1K accuracy) does not imply a better representation for generation; across encoders, families, and controlled interventions, global performance fails to predict REPA effectiveness.*

3 SPATIAL STRUCTURE MATTERS MORE

We advance and rigorously test a single hypothesis: it is the spatial structure of a target representation, rather than its global information, that governs generation performance. To this end, we first formalize a family of straightforward metrics that quantify the spatial self-similarity structure (Shechtman & Irani, 2007) of target representations. We then establish two results that jointly confirm the hypothesis: every spatial structure metric correlates far more strongly with gFID than ImageNet-1K accuracy does, and these same metrics account, in a unified way, for each of the previously unexplained observations of §2.

Measuring spatial self-similarity structure. Given an image $\mathcal{I}$ and vision encoder $\mathcal{E}$, we define:

- *Self-similarity.* Let $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$ be the extract patch representations, with $T = H \times W$ patches. Let kernel $K_{\mathcal{X}}(\cdot, \cdot) \in \mathbb{R}$ measure appearance self-similarity (Shechtman & Irani, 2007) between patch tokens. Here, we use the cosine kernel $K_{\mathcal{X}}(t, t') = \frac{\langle \mathbf{x}_t, \mathbf{x}_{t'} \rangle}{\|\mathbf{x}_t\|_2 \|\mathbf{x}_{t'}\|_2}$.
- *Spatial distance.* Let $\mathcal{P} \in \mathbb{N}^{T \times 2}$ be the spatial location of each of the T tokens in coordinate space and $d(\cdot, \cdot) \in \mathbb{N}$ be the Manhattan distance between pairs of tokens.

We then define the *spatial* self-similarity metric as a measure of how the self-similarity $K_{\mathcal{X}}$ varies with the lattice distance d between patch tokens. Intuitively, larger values indicate stronger spatial organization, with spatially closer patches more similar to one another than distant ones. By default, we adopt a simple correlogram-contrast (local vs. distant) metric (Huang et al., 1997):

$$\text{LDS}(\mathcal{X}, \mathcal{P}) = \mathbb{E}_{d(t, t') \in (0, r_{\text{near}})} K_{\mathcal{X}}(t, t') - \mathbb{E}_{d(t, t') \geq r_{\text{far}}} K_{\mathcal{X}}(t, t').$$

The final spatial self-similarity score (LDS) is obtained as the expectation over patch representations $\mathcal{X}$. We set $r_{\text{near}} = r_{\text{far}} = H/2$ throughout; the resulting correlation is robust to their exact choice. Appendix B provides the full details along with three alternative spatial metrics (CDS, SRSS, RMSC), each of which is equally predictive (see Fig. 4), confirming that the finding does not hinge on any single formulation.

Spatial structure correlates far more strongly with generation performance than linear probing.

We conduct a large-scale correlation analysis across 27 diverse vision encoders. As Figure 4 shows, the conventionally used linear probing accuracy exhibits only a very weak association with generation FID (Pearson $|r| = 0.26$), whereas every SSM metric correlates far more strongly with generation performance (Pearson $|r| > 0.85$)—a more than threefold increase in predictive power that establishes spatial structure, and not global accuracy, as the reliable predictor of REPA quality.

Generalization across diffusion model scales. The correlation holds uniformly across model scales (SiT-B, SiT-L, SiT-XL), as verified in Figure 5. Linear probing remains weakly predictive at every scale ($|r| < 0.306$), while spatial structure sustains a much stronger correlation with generation performance throughout ($|r| > 0.826$), confirming that the predictive advantage of spatial structure is scale-invariant rather than an artifact of any particular model size.

- **Finding 2.** *Spatial structure correlates far more strongly with generation performance than linear probing accuracy, and the same metrics account for trends that global performance leaves unexplained.*

Spatial structure explains previously unexplained trends. As established in §2, global performance (ImageNet-1K accuracy) is not a predictive measure of effectiveness for representation alignment. The spatial structure metrics resolve each of those anomalies, providing a single, consistent account of trends that the accuracy-centric view leaves unexplained.

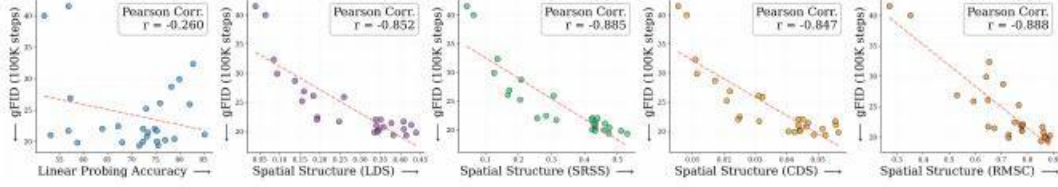


Figure 4: **Spatial structure correlates far more strongly with generation quality than linear probing.** Correlation analysis across 27 diverse vision encoders using SiT-XL/2 and REPA. Linear probing correlates only weakly with FID (Pearson $|r| = 0.260$), whereas every spatial structure metric—LDS ($|r| = 0.852$), SRSS ($|r| = 0.885$), CDS ($|r| = 0.847$), and RMSC ($|r| = 0.888$)—correlates far more strongly with generation performance, a more than threefold advantage that holds across all four metric definitions. See Fig. 10 for detailed plots with encoder labels.

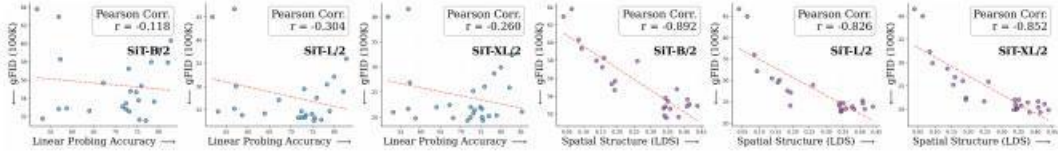


Figure 5: **Correlation analysis across model scales.** At every diffusion model scale (SiT-B, SiT-L, SiT-XL), spatial structure (right) correlates consistently more strongly with gFID than linear probing (left) does, establishing that the predictive advantage of spatial structure is scale-invariant rather than an artifact of any single model size.

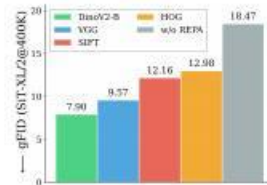
(1) *PE-Spatial-B vs. PE-Core-G.* Figure 2 shows that PE-Core-G achieves far higher ImageNet-1K accuracy (82.8% vs. 53.1%) yet performs worse as the REPA target representation (FID 32.3 vs. 22.0). Spatial structure resolves the paradox: despite its lower global performance, PE-Spatial-B exhibits substantially stronger spatial structure than PE-Core-G, which directly accounts for its better generation.

(2) *SAM2 outperforms “better” vision encoders.* §2 shows that although SAM2 attains much lower ImageNet-1K accuracy (24.1%), it yields better generation than encoders with $\sim 60\%$ higher accuracy (Fig. 3a). These gains follow directly from SAM2’s stronger spatial structure.

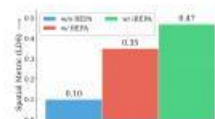
(3) *Larger models within the same encoder family underperform.* As shown in Fig. 3b, larger models within an encoder family attain higher ImageNet-1K accuracy yet can carry weaker spatial structure, and consequently generate worse under REPA. This mechanism also subsumes the observation of Yu et al. (2024) that larger models can yield similar or worse generation, replacing their distillation-based conjecture with a measurable property.

(4) *Adding global information to patch tokens via the CLS token hurts generation.* In Figure 3c, mixing the CLS token into patch tokens increases global performance but reduces the spatial contrast among patch tokens, so that patch tokens become highly similar to otherwise unrelated tokens (e.g., foreground object to background). The resulting loss of spatial structure is precisely what degrades generation.

If spatial structure matters more, can classical SIFT or HOG features serve as REPA targets? Yes. Although they are certainly not the strongest targets, classical spatial descriptors such as SIFT (Lowe, 1999) and HOG (Dalal & Triggs, 2005), together with intermediate VGG features (Simonyan & Zisserman, 2014), all yield performance gains under REPA. That representations devoid of learned global semantics can still improve generation is direct evidence that representation alignment benefits from spatial structure alone, without any additional global information.



Can spatial metrics explain the gains obtained with REPA? Yes. The proposed spatial metrics account not only for the gains of standard REPA but also for the further improvements delivered by our enhanced training recipe (iREPA), which we introduce next in §4.



4 iREPA: IMPROVING REPRESENTATION ALIGNMENT BY ACCENTUATING WHAT MATTERS

Having established that spatial structure governs the generative utility of a target representation, we now operationalize this principle directly. We introduce two minimal, architecturally targeted modifications to the original REPA training recipe, each designed to accentuate the transfer of spatial structure from the teacher (the pretrained vision encoder) to the student (the diffusion transformer). Together they constitute iREPA, a recipe that requires fewer than four lines of code yet consistently and substantially improves convergence.

Convolutional projection layer in place of the MLP. Standard REPA employs a three-layer MLP projection to map the diffusion feature dimensionality onto that of the external representation. Because this projection is applied to each patch token independently, it necessarily discards the spatial relationships among tokens: as Figure 6a demonstrates, the MLP mapping is lossy and measurably attenuates the spatial contrast between patch tokens. We therefore replace it with a lightweight convolutional projection (kernel size 3, padding 1) that operates directly over the spatial grid. The locality of the convolution imposes, by construction, an inductive bias that preserves the neighborhood relationships among patch tokens—precisely the property our analysis identifies as decisive for generation.

Spatial normalization layer. Consistent with Siméoni et al. (2025), the patch tokens of pretrained vision encoders carry a substantial global component, evidenced by the high linear probing accuracy obtained from the mean of the patch tokens (Figure 14). Figure 6b makes the underlying trade-off explicit: mixing this global component (the mean of the patch tokens) into the individual tokens raises global performance but erodes the spatial contrast between tokens, so that unrelated tokens (e.g., a foreground object and the background) come to appear spuriously similar.

The mechanistic account of §3 yields a direct prescription: deliberately suppressing this global component should sharpen the spatial contrast among patch tokens and thereby strengthen the pairwise-similarity signal that drives REPA. We therefore apply a simple spatial normalization layer (Ulyanov et al., 2016) to the patch tokens of the target representation:

$$\mathbf{y} = \frac{\mathbf{x} - \gamma \mathbb{E}[\mathbf{x}]}{\sqrt{\text{Var}[\mathbf{x}] + \epsilon}},$$

where $\mathbf{x} \in \mathbb{R}^{B \times T \times D}$ denotes the patch tokens, the expectation and variance are computed across the spatial dimension, γ controls the fraction of the global component that is removed, and $\epsilon = 10^{-6}$ ensures numerical stability. Setting γ to modulate the removed global component thus provides a direct, single-parameter control over the strength of the spatial contrast supplied to the alignment objective.

Algorithm 1 Summary of key iREPA modifications to standard REPA training

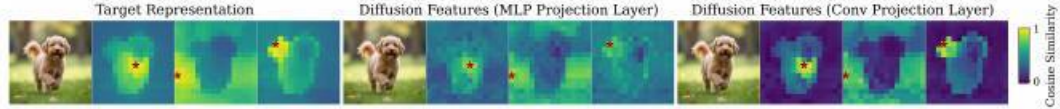
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# 1. Conv projection instead of MLP
proj_layer = nn.Conv2d(D_in, D_out, kernel_size=3, padding=1)

# 2. Spatial normalization on encoder features [B, T, D]
x = x - gamma * x.mean(dim=1, keepdim=True)
x = x / (x.std(dim=1, keepdim=True) + 1e-6)
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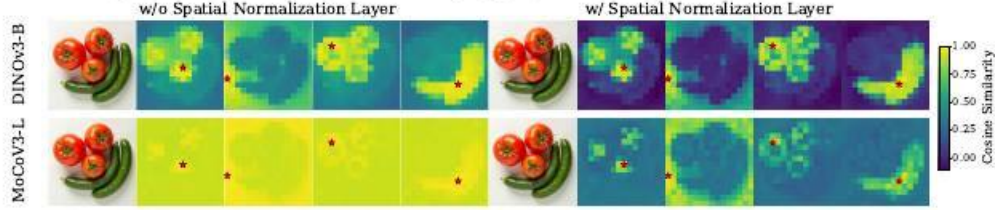
4.1 EXPERIMENTS

We validate iREPA through an extensive suite of experiments on ImageNet 256×256 , organized around three research questions that together establish its effectiveness, its scalability, and the generality of the underlying principle:

- Does iREPA consistently accelerate the convergence of diffusion transformers over REPA across a diverse set of external representations? (Figure 7, 8, 12, 13, 15)
- Does iREPA scale with model size and generalize across variations in training configuration? (Table 1 [a,b,c], 2, 4, 5, 6, and, Figure 12, 16)



(a) *Simpler projection layer for REPA.* Because it acts on each patch token independently, the standard MLP projection layer in REPA (middle) discards spatial information when transferring features from the target representation (left) to the diffusion features. A simpler convolutional layer, which operates over the spatial grid, preserves this spatial information far more faithfully (right).



(b) *Spatial normalization layer.* The patch tokens of pretrained vision encoders carry a global component that suppresses spatial contrast, causing tokens in one semantic region (e.g., tomato) to exhibit substantial cosine similarity with unrelated tokens (e.g., background or cucumber). Suppressing this global component sharpens the spatial contrast among patch tokens and thereby strengthens the pairwise-similarity signal that drives generation.



(c) *Overall impact of the improved training recipe (iREPA) on the spatial structure of the diffusion features.*

Figure 6: Motivation for the two minimal modifications (iREPA) that accentuate the transfer of spatial features from the target representation to the diffusion features: each modification targets a distinct source of spatial information loss—the token-independent MLP projection (a) and the global overlay in the target representation (b)—and their joint effect on the diffusion features is shown in (c). All results reported with SiTL/2 and REPA at 100K iterations.

- Does iREPA generalize to more recent representation alignment methods such as REPA-E (Leng et al., 2025a) and MeanFlow w/ REPA (Geng et al., 2025), and to pixel-space diffusion models such as JiT w/ REPA (Li & He, 2025b)? (Table 3 [a,b])

Convergence Speed. We evaluate the convergence behavior of iREPA across a diverse set of vision encoders (DINOv3-B, WebSSL-1B, PE-Core-G, CLIP-L, MoCov3, PE-Lang-G) and two diffusion model sizes (SiT-XL/2 and SiT-B/2), reported in Fig. 7. Across every combination of target representation and model size, iREPA converges faster than baseline REPA—without exception—establishing that the benefit of accentuating spatial transfer is uniform rather than encoder- or scale-specific.

Target representation. We assess the generality of iREPA across vision encoders in Fig. 8 and Table 4. iREPA improves generation quality for every one of the vision encoders considered, with the complete comparison across all 27 encoders reported in Appendix C, E. The uniformity of the gain across this breadth of encoders confirms that spatial transfer, and not any encoder-specific property, is what iREPA exploits.

Enc. Size	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Pr. $\uparrow$	Rc. $\uparrow$
PE-B (90M)	59.6	22.5	6.23	0.63	0.60
+iREPA	73.3	17.5	6.04	0.66	0.62
PE-L (320M)	48.6	28.7	6.53	0.59	0.60
+iREPA	76.3	17.6	6.36	0.65	0.61
PE-G (1.88B)	42.7	32.3	6.70	0.57	0.59
+iREPA	70.8	19.5	6.35	0.64	0.61

(a) Vision Encoder size

Model Size	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Pr. $\uparrow$	Rc. $\uparrow$
SiT-B	27.5	49.50	7.00	0.46	0.59
+iREPA	34.1	43.37	7.87	0.50	0.60
SiT-L	55.7	24.10	6.25	0.62	0.60
+iREPA	66.9	20.28	6.14	0.63	0.62
SiT-XL	70.3	19.06	5.83	0.65	0.61
+iREPA	77.9	16.96	6.26	0.66	0.61

(b) Model size

Aln. Depth	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Pr. $\uparrow$	Rc. $\uparrow$
Layer 4	28.1	49.0	6.83	0.46	0.60
+iREPA	41.7	36.0	6.61	0.52	0.62
Layer 6	26.9	50.6	7.00	0.46	0.59
+iREPA	42.4	35.3	6.90	0.53	0.62
Layer 8	24.7	54.8	7.35	0.44	0.58
+iREPA	38.3	38.9	7.19	0.51	0.62

(c) Encoder depth (SiT-B/2)

Table 1: **Variation in training settings.** iREPA delivers consistent gains over baseline REPA across encoder size (a), diffusion model size (b), and alignment depth (c); notably, the relative gain grows with both encoder capacity and diffusion model scale, exactly the regimes in which spatial structure is most at risk. Unless otherwise specified, all results use DINOv2 as the encoder, SiT-XL/2, 100K steps, and vanilla REPA as the baseline.

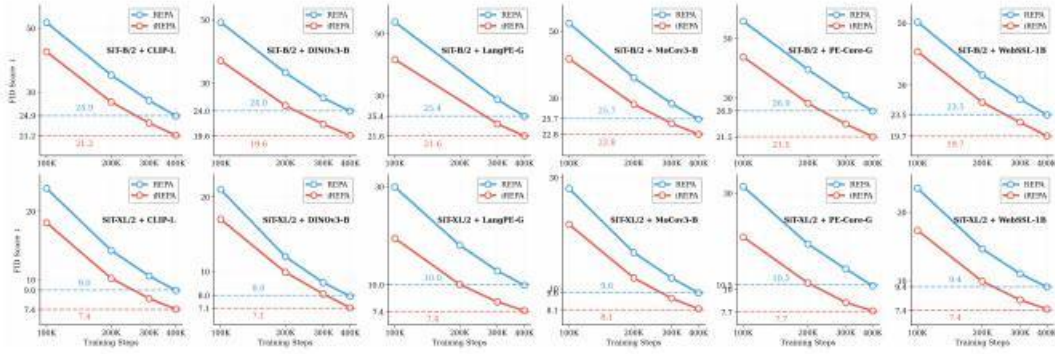


Figure 7: **Accentuating spatial features consistently improves convergence speed.** FID convergence curves show that iREPA accelerates convergence over baseline REPA across every vision encoder and diffusion model size considered, without exception.

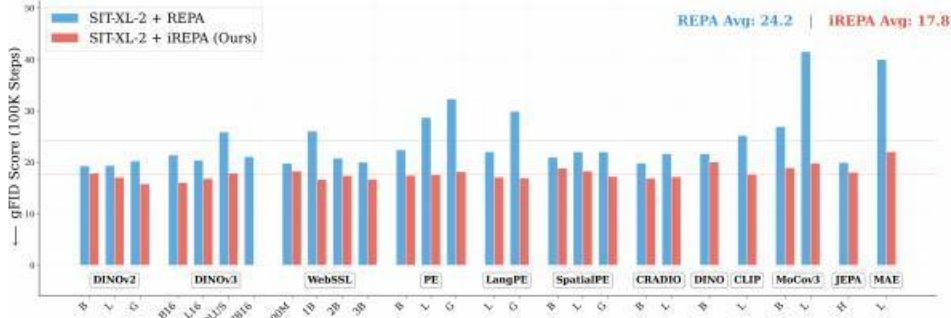


Figure 8: **Variation in target representation.** Across all 27 vision encoders, accentuating the transfer of spatial features from the target representation to the diffusion features (iREPA) yields consistent improvements in convergence speed—the uniformity confirming that the gain reflects the spatial-transfer principle rather than any encoder-specific property. See Appendix C for further results across diverse settings.

Encoder size. Table 1a reports how iREPA generalizes across encoder sizes within the PE family, spanning PE-B (90M), PE-L (320M), and PE-G (1.88B). iREPA improves performance at every encoder size, and, tellingly, the relative improvement grows monotonically with encoder capacity (22.2% for PE-B, 38.8% for PE-L, and 39.6% for PE-G). This is exactly the regime in which larger encoders tend to shed spatial structure, and therefore the regime in which restoring it is most valuable—so the trend is itself a direct corroboration of the spatial-structure account.

Scalability. Table 1b examines how iREPA scales with the size of the diffusion model. The spatial modifications improve performance at every model scale and, moreover, yield progressively larger relative gains as the model grows—establishing that the benefit of accentuating spatial transfer scales favorably with diffusion model size rather than diminishing, a property essential for its use in larger-scale generative training.

Encoder depth. Table 1c reports iREPA across alignment depths, with all results obtained on SiT-B/2 at 100K iterations using DINOv3-B as the target representation. iREPA improves consistently over baseline REPA at every alignment depth, confirming that the gains are intrinsic to the spatial modifications rather than contingent on a particular choice of alignment layer.

Ablation on individual components. We isolate the contribution of each iREPA component in Table 2. The spatial normalization layer and the convolutional projection layer each improve generation quality over baseline REPA on their own, and the two prove complementary: applying them jointly yields the best result across every encoder. This complementarity confirms that the two

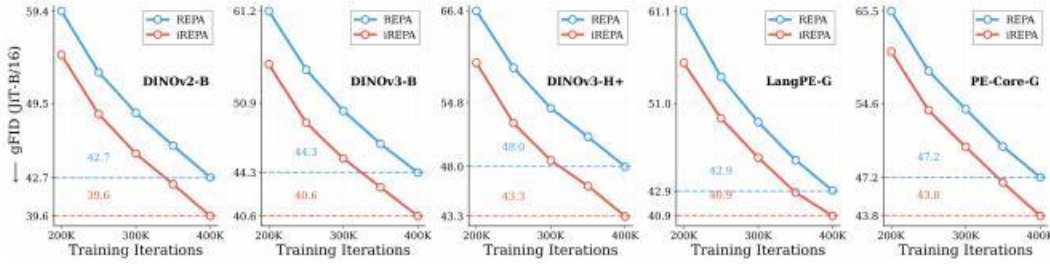


Figure 9: **Convergence comparison with pixel-space diffusion (JiT).** FID convergence curves comparing REPA and iREPA with pixel-space JiT (Li & He, 2025b) across vision encoders. Accentuating the transfer of spatial structure again yields consistently faster convergence across encoders, establishing that the principle carries over intact from latent-space to pixel-space diffusion. All results are reported with JiTB/16 (Li & He, 2025b), batch size 256, and without classifier-free guidance. See Table 10 for further results.

Method	DiNOv2-B			DiNOv3-B			WebSSL-1B			PE-Core-G		
	FID↓	IS↑	sFID↓	FID↓	IS↑	sFID↓	FID↓	IS↑	sFID↓	FID↓	IS↑	sFID↓
Baseline REPA	19.06	70.3	5.83	21.47	63.4	6.19	26.10	53.0	6.90	32.35	42.7	6.70
iREPA (w/o spatial norm)	18.52	73.3	6.11	17.76	74.7	5.81	21.17	64.6	6.27	24.97	57.4	6.21
iREPA (w/o conv proj)	17.66	72.8	6.03	18.28	70.8	6.18	18.44	71.0	6.22	21.72	61.5	6.26
iREPA (full)	16.96	77.9	6.26	16.26	78.8	6.14	16.66	77.5	6.18	18.19	75.0	6.03

Table 2: **Ablation on individual components.** Across every encoder, the spatial normalization layer and the convolutional projection layer (§4) each deliver substantial convergence gains over baseline REPA on their own; the two are complementary, and combining them yields the best result in every column. All results reported at 100K steps with SiT-XL/2.

modifications address distinct sources of spatial information loss—the global overlay in the target representation and the token-independent MLP mapping, respectively.

Training recipe. Finally, we test whether iREPA transfers across representation alignment recipes, evaluating it on REPA-E (Leng et al., 2025a) and MeanFlow w/ REPA (Geng et al., 2025). As Table 3 shows, the spatial modifications deliver convergence gains under both recipes, establishing that accentuating spatial transfer is a recipe-agnostic principle rather than an artifact of the original REPA setup.

Classifier-free guidance. Table 4 reports generation quality with classifier-free guidance across vision encoders at 400K training iterations. iREPA improves generation quality for every encoder both with and without classifier-free guidance, confirming that its gains persist under the guidance regime used at inference and are not washed out by it.

Pixel-space diffusion (JiT). We further extend iREPA to pixel-space diffusion models, evaluating it on JiT-B (Li & He, 2025b). Figure 9 contrasts REPA and iREPA under JiT (Li & He, 2025b): accentuating the transfer of spatial information again yields consistently faster convergence across vision encoders (e.g., DiNOv2, DiNOv3, PE etc.), demonstrating that the principle carries over intact from latent-space to pixel-space diffusion, and thus is not specific to the latent-diffusion setting in which it was first identified.

- **Finding 3.** Accentuating the transfer of spatial structure from the target representation to the diffusion features consistently boosts convergence speed under representation alignment, across encoders, model sizes, and training recipes (REPA, REPA-E, MeanFlow, and JiT).

5 RELATED WORK

We situate our contribution among the most directly relevant lines of work here, and defer an extended discussion to Appendix H.

Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
WebSSL-1B	52.8	26.5	5.20	0.620	0.585
+iREPA-E	87.0	13.2	5.28	0.699	0.598
PE-G	50.9	25.9	5.68	0.612	0.576
+iREPA-E	80.0	16.4	5.40	0.667	0.616
DINOv3-B	82.2	14.4	4.68	0.694	0.596
+iREPA-E	93.6	11.7	4.57	0.703	0.613
DINOv2-B	87.5	12.9	5.40	0.708	0.586
+iREPA-E	91.3	12.1	4.86	0.712	0.602

(a) REPA-E (SiT-XL/2)

Encoder	w/o CFG				w/ CFG			
	4 NFE		1 NFE		4 NFE		1 NFE	
	IS $\uparrow$	FID $\downarrow$	IS $\uparrow$	FID $\downarrow$	IS $\uparrow$	FID $\downarrow$	IS $\uparrow$	FID $\downarrow$
WebSSL-1B	27.22	51.40	24.14	58.67	87.85	16.59	69.06	23.74
+iREPA	31.48	45.67	27.33	55.70	100.74	13.89	78.67	20.69
DINOv3-B	28.36	49.58	25.47	57.01	93.30	15.56	72.41	22.55
+iREPA	33.63	44.52	29.67	53.75	124.54	11.05	98.93	17.32
DINOv2-B	28.77	48.87	25.43	56.63	94.40	15.28	71.95	22.17
+iREPA	33.63	44.52	29.67	53.75	111.44	12.59	90.25	18.62

(b) MeanFlow w/ REPA (SiT-B/2)

Table 3: **Variation in training recipes.** Applied on top of REPA-E (a) and MeanFlow with REPA (b), our spatial modifications yield consistent gains under both recipes, establishing that accentuating spatial transfer is recipe-agnostic rather than specific to the original REPA setup. For MeanFlow, we report results at 1 and 4 NFEs, using a CFG value of 2.0. All results are reported at 100K training iterations.

Vision Encoder	Steps	w/o CFG					w/ CFG				
		IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
DINOv2-B	100K	69.20	19.3	5.89	0.64	0.61	157.2	6.35	5.91	0.769	0.536
+iREPA	100K	77.92	16.9	6.26	0.66	0.61	179.3	5.15	6.23	0.783	0.544
DINOv2-B	400K	127.4	7.76	5.06	0.70	0.66	263.0	1.98	4.60	0.799	0.610
+iREPA	400K	128.6	7.52	4.89	0.71	0.65	268.8	1.93	4.59	0.799	0.600
DINOv3-B	100K	63.64	21.4	6.14	0.63	0.60	144.0	7.57	6.09	0.762	0.526
+iREPA	100K	78.79	16.2	6.14	0.66	0.61	181.9	4.87	6.10	0.780	0.547
DINOv3-B	400K	126.7	8.10	5.06	0.70	0.66	261.2	1.99	4.58	0.799	0.609
+iREPA	400K	132.9	7.13	4.93	0.71	0.66	272.4	1.89	4.58	0.799	0.600
WebSSL-1B	100K	53.87	25.5	6.57	0.61	0.59	124.0	9.59	6.37	0.756	0.515
+iREPA	100K	77.47	16.6	6.18	0.66	0.61	177.6	5.09	6.11	0.787	0.538
WebSSL-1B	400K	116.9	9.39	5.14	0.70	0.64	250.5	2.24	4.61	0.809	0.580
+iREPA	400K	130.8	7.48	4.91	0.70	0.65	271.6	1.90	4.58	0.798	0.609
PE-Core-G	100K	42.74	32.3	6.70	0.57	0.59	97.2	14.1	6.56	0.714	0.525
+iREPA	100K	75.01	18.1	6.03	0.64	0.61	176.8	5.66	6.08	0.771	0.544
PE-Core-G	400K	109.4	10.4	5.00	0.69	0.64	238.4	2.44	4.57	0.805	0.585
+iREPA	400K	132.0	7.78	5.02	0.70	0.65	275.4	1.93	4.59	0.796	0.606

Table 4: **Results across encoders, with and without classifier-free guidance.** With SiT-XL/2 as the base model, iREPA improves generation quality for every encoder under both guidance settings and at both 100K and 400K steps—without exception. See Appendix E for detailed results on SiT-B/2 & SiT-L/2 across all encoders.

Representation alignment for generation. A growing body of work exploits external representations to improve the training of diffusion models (Pernias et al., 2023; Fuest et al., 2024). In particular, (Yu et al., 2024; Yao & Wang, 2025; Leng et al., 2025a;b; Kouzelis et al., 2025) demonstrate that aligning the internal features of a diffusion model with the clean image features of a pretrained vision encoder yields substantial performance gains, and the idea has since been extended to video generation (Zhang et al., 2025) and 3D generation (Wu et al., 2025), as well as to the training of unified models (Ma et al., 2025a). Across these successes, however, the precise mechanism by which self-supervised features enhance diffusion training has remained unresolved. We close this gap: we identify the property of the target representation that actually drives generation—its spatial structure—and translate that understanding into a concrete, improved training recipe.

Spatial vs. global information trade-off in pretrained vision encoders. A parallel line of work investigates the tension between global and spatial information in pretrained vision encoders (Bolya et al., 2025; Siméoni et al., 2025). (Siméoni et al., 2025) show that continued self-supervised training can increase the similarity between the global CLS token and the patch tokens, degrading performance on dense spatial tasks, while (Bolya et al., 2025; Heinrich et al., 2024) explicitly train spatially tuned models for such tasks. Building on this trade-off, we establish that for generation it is the spatial structure of a vision encoder, and not its global information, that is decisive—a finding that reframes how external representations should be selected and trained for generative modeling.

6 CONCLUSION

This paper resolves a fundamental question about representation alignment: which property of the target representation actually drives its effectiveness—global information or spatial structure? Through a large-scale empirical analysis spanning 27 vision encoders and three diffusion model scales, together with controlled interventions, we establish that it is spatial structure, and not global information, that governs the effectiveness of representation alignment, and we introduce fast-to-compute spatial metrics that make this dependence quantitatively predictive. We then operationalize the principle with two minimal modifications that accentuate the transfer of spatial information from the target representation to the diffusion features. The resulting method, iREPA, consistently accelerates the convergence of REPA across diverse vision encoders, diffusion model sizes, and training recipes (REPA, REPA-E, MeanFlow, and JiT). By reframing the working mechanism of representation alignment, these findings chart a concrete path toward selecting and training external representations far more effectively for generative modeling.

REPRODUCIBILITY STATEMENT

We provide all implementation details and hyperparameters in Appendix G. Our code, model checkpoints, and analysis results will be released upon acceptance.

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A DETAILED CORRELATION ANALYSIS

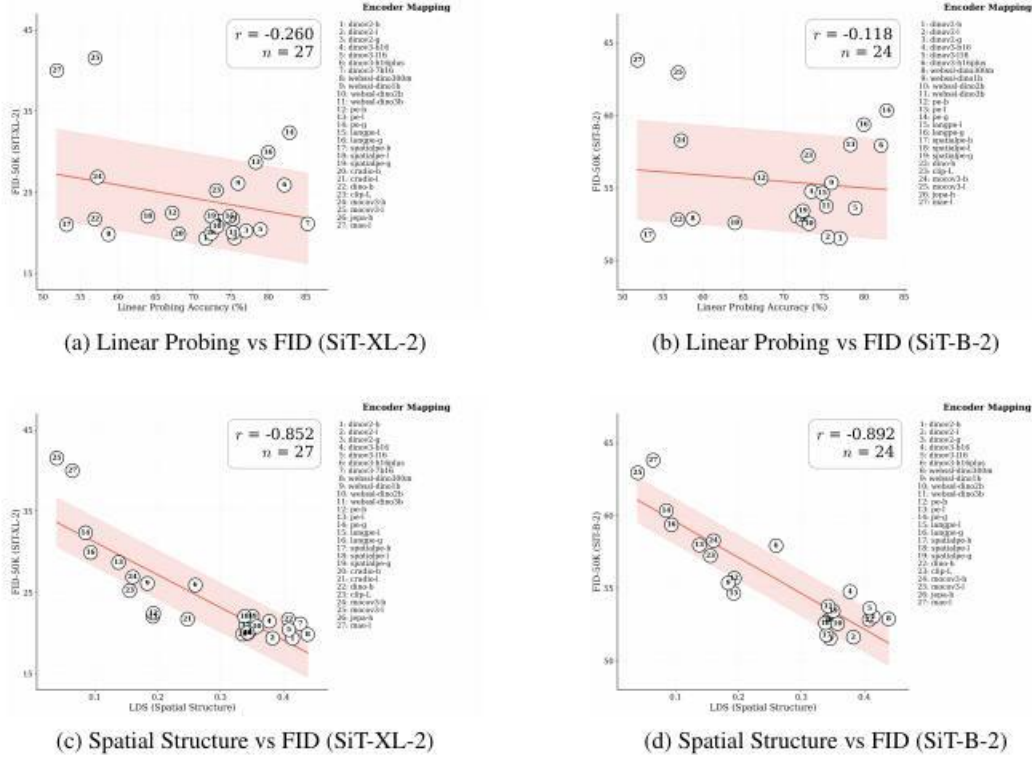


Figure 10: Detailed correlation analysis with encoder labels. The top row plots linear probing accuracy against FID for the SiT-XL-2 and SiT-B-2 models, and the bottom row plots spatial structure (LDS) against FID; each point is labeled with its encoder number (see legend). At both model scales, the spatial structure metric correlates substantially more strongly with generation quality than linear probing accuracy does, with the per-encoder labels making the tightness of the spatial-structure fit and the scatter of the linear-probing fit directly visible.

B SPATIAL SELF-SIMILARITY METRICS

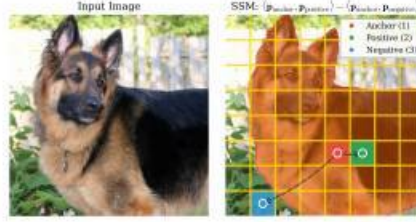


Figure 11: **Explaining the semantic-region self-similarity (SRSS) metric.** Visual explanation of the SRSS metric: for each image we compute a mask with SAM2 and sample anchor–positive–negative triplets. The SSM is the difference in cosine similarity between anchor–positive pairs (within the mask) and anchor–negative pairs (outside the mask). Intuitively, larger values indicate that patches on the same semantic region are more similar to one another than to patches on unrelated regions—a direct signature of preserved spatial structure.

Setup. Given an image $\mathcal{I}$ and vision encoder $\mathcal{E}$, we extract patch tokens $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$ with $T = H \times W$ and indices $t \in [T]$ placed on an $H \times W$ lattice. Let $d : [T] \times [T] \rightarrow \mathbb{N}$ be the Manhattan distance, and let the (cosine) self-similarity kernel be

$$K_{\mathcal{X}}(t, t') = \frac{\langle \mathbf{x}_t, \mathbf{x}_{t'} \rangle}{\|\mathbf{x}_t\|_2 \|\mathbf{x}_{t'}\|_2} \in [-1, 1],$$

a standard local self-similarity measure (Shechtman & Irani, 2007).

- *Patch token representation:* $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$ where $T = H \times W$ is the spatial grid of patches (patch index set $[T]$). Also let $d : [T] \times [T] \rightarrow \mathbb{N}$ be Manhattan distance on the $H \times W$ grid.
- *Self-similarity kernel:* $K : [T] \times [T] \rightarrow \mathbb{R}$ measuring self-similarity (Shechtman & Irani, 2007) between patch tokens. We use the cosine kernel $K_{\mathcal{X}}(t, t') = \langle \mathbf{x}_t, \mathbf{x}_{t'} \rangle / (\|\mathbf{x}_t\|_2 \|\mathbf{x}_{t'}\|_2)$.
- *Spatial self-similarity metric:* a functional $m : \mathcal{K} \times \mathcal{D} \rightarrow \mathbb{R}$ measures how self-similarity K between patch tokens varies with lattice distance d . Intuitively, larger values indicate stronger spatial organization (near patches more similar to each other than far away patches).

We now define several straightforward metrics that quantify the spatial self-similarity structure of a target representation. Each is a low-cost functional of the self-similarity kernel and the lattice distance, and, as our correlation analysis shows, each is far more predictive of generation quality than global accuracy.

Local vs. Distant Similarity (LDS). We first adopt a simple correlogram-contrast (local vs. distant) metric (Huang et al., 1997):

$$\text{LDS}(\mathcal{X}) = \mathbb{E}[K_{\mathcal{X}}(t, t') \mid d(t, t') < r_{\text{near}}] - \mathbb{E}[K_{\mathcal{X}}(t, t') \mid d(t, t') \geq r_{\text{far}}].$$

where r_{near} and r_{far} are hyperparameters, set by default to $r_{\text{near}} = r_{\text{far}} = H/2$; the resulting correlation is robust to their exact values. A larger LDS indicates that, *on average*, spatially closer patches are more similar than distant ones—a direct signature of preserved spatial structure.

Correlation Decay Slope (CDS). Given the patch tokens $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$, we compute the spatial correlogram $g_{\mathcal{X}}(\delta) = \mathbb{E}[K_{\mathcal{X}}(t, t') \mid d(t, t') = \delta]$ over distances $\delta \in \{1, \dots, \Delta\}$, fit a least-squares line $\hat{g}_{\mathcal{X}}(\delta) \approx \alpha + \beta\delta$ to it, and define:

$$\text{CDS}(\mathcal{X}) = -\hat{\beta}$$

where $\hat{\beta}$ is the fitted slope. A larger CDS corresponds to faster decay of similarity with distance, and hence to stronger spatial organization.

Semantic-Region Self-Similarity (SRSS). For each image, we sample a binary mask $\mathbf{M} \in \mathbb{R}^{H \times W}$ enclosing a random object using SAM2 (Ravi et al., 2024), downsampled to $H \times W$. From this mask we draw three classes of points—anchor, positive, and negative—where the anchor and positive lie within the mask (and are therefore semantically related) and the negative lies outside it (and is therefore less related to the anchor). An encoder that preserves spatial structure should assign high

cosine similarity to the anchor–positive pair and low similarity to the anchor–negative pair. We accordingly define the Spatial Structure Metric (SSM) as:

$$\text{SSM}(\mathbf{P}) = \mathbb{E}_{\text{anchor} \in \mathbf{M}} [\cos(\mathbf{p}_{\text{anchor}}, \mathbf{p}_{\text{pos}}) - \cos(\mathbf{p}_{\text{anchor}}, \mathbf{p}_{\text{neg}})]$$

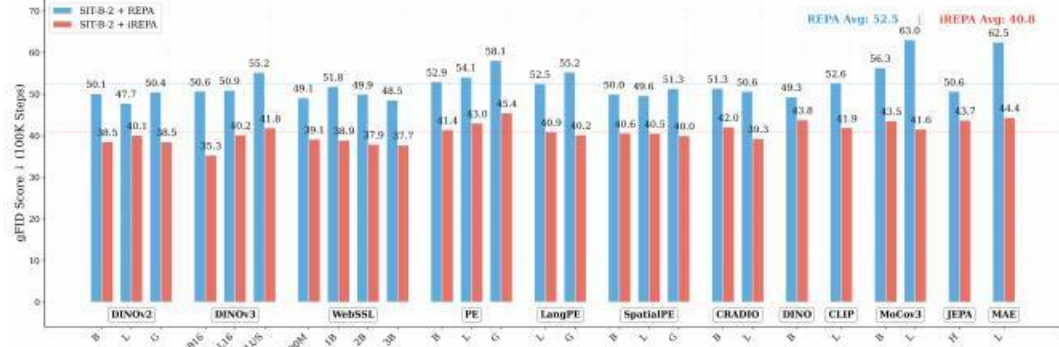
where positive points lie within Manhattan distance $d \leq r_{\text{near}}$ of the anchor inside the mask, and negative points lie at distance $d \geq r_{\text{far}}$ outside the mask.

RMS Spatial Contrast (RMSC). Finally, we introduce a simple contrast metric that captures the dispersion among patch token representations. Given ℓ_2 -normalized features $\hat{\mathbf{x}}_t = \mathbf{x}_t / \|\mathbf{x}_t\|_2$ for each patch t , we compute:

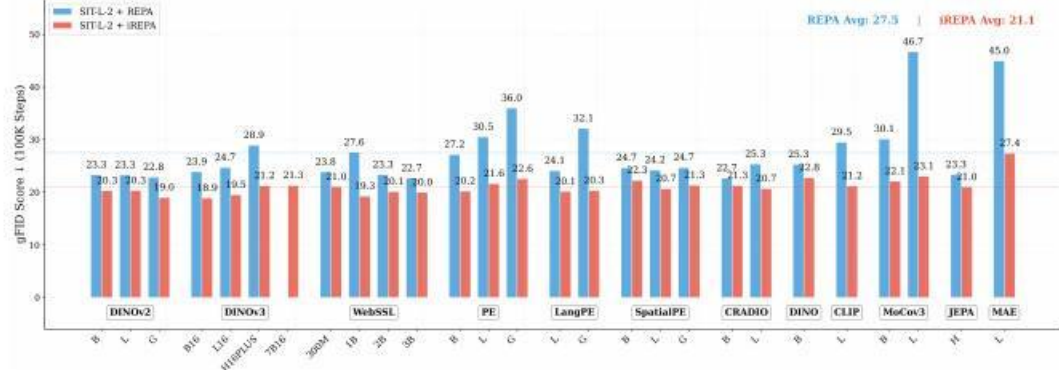
$$\text{RMSC}(\mathcal{X}) = \sqrt{\frac{1}{T} \sum_{t=1}^T \|\hat{\mathbf{x}}_t - \bar{\mathbf{x}}\|_2^2}$$

where $\bar{\mathbf{x}} = \frac{1}{T} \sum_{t=1}^T \hat{\mathbf{x}}_t$ is the mean normalized feature across all spatial locations. A higher RMSC reflects greater spatial diversity among features, and hence preserved spatial structure; a lower RMSC reflects a more uniform feature distribution—the hallmark of the loss of spatial structure that occurs when a dominant global component overwhelms it (Siméoni et al., 2025).

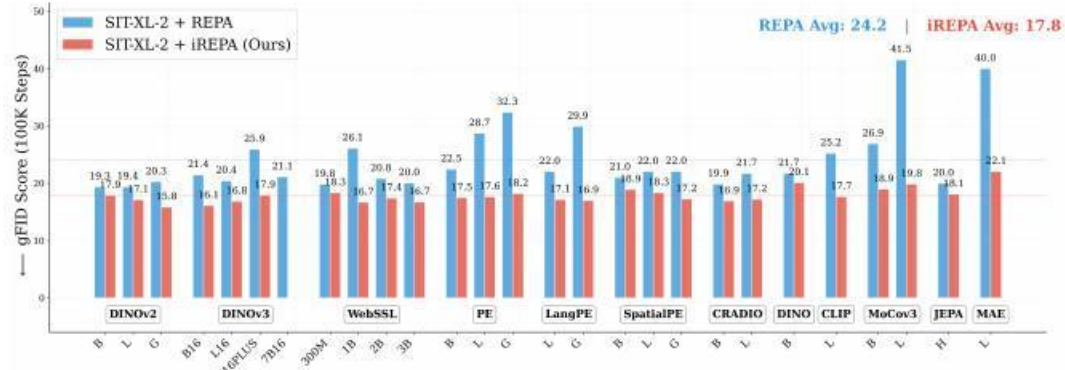
C VISION ENCODER VARIATION



(a) SiT-B-2 (130M parameters)



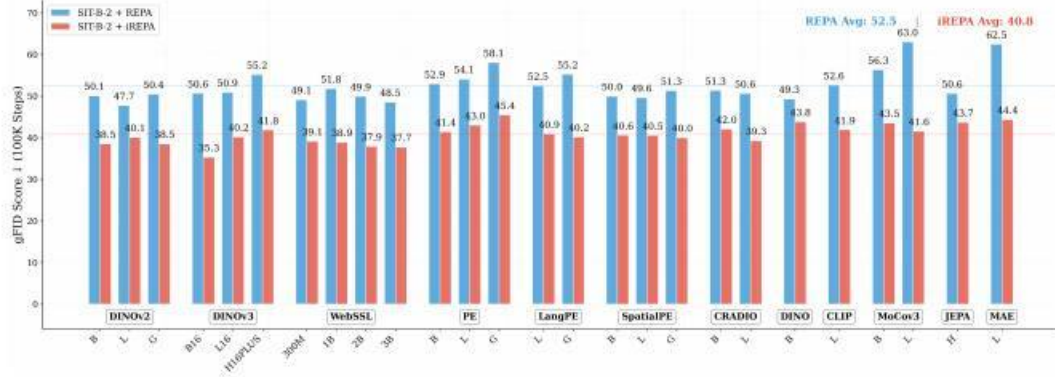
(b) SiT-L-2 (458M parameters)



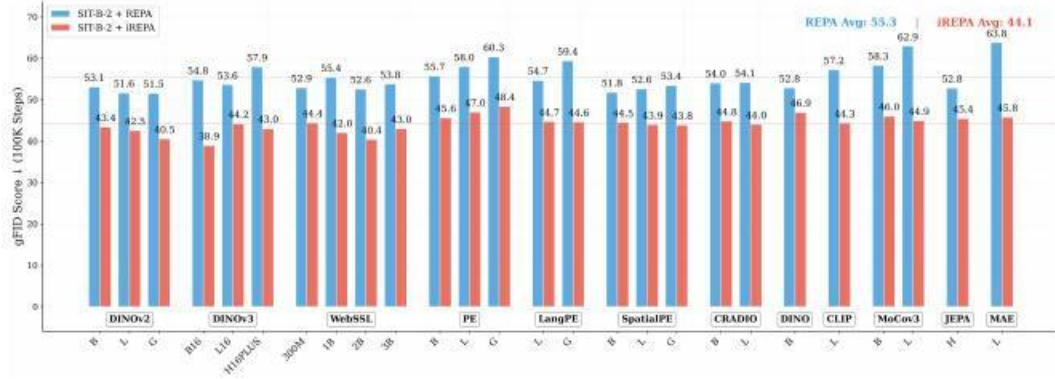
(c) SiT-XL-2 (675M parameters)

Figure 12: **Variation in vision encoders.** Ablation across vision encoders for the SiT-B-2, SiT-L-2, and SiT-XL-2 models. iREPA improves generation quality for every encoder at every model size, confirming that the benefit is uniform rather than encoder- or scale-specific.

D ENCODER DEPTH VARIATION



(a) SiT-B-2 with 6 encoder layers



(b) SiT-B-2 with 8 encoder layers

Figure 13: **Effect of encoder depth on iREPA performance.** SiT-B-2 performance at two encoder depths (6 vs. 8 layers) across a range of vision encoders. iREPA improves consistently at both depths—slightly more so at 8 layers—and both configurations exhibit substantial FID gains once spatial regularization is applied, confirming that the benefit is intrinsic to the spatial modifications rather than tied to a particular depth.

E COMPREHENSIVE RESULTS ACROSS VISION ENCODERS

E.1 SiT-B/2 RESULTS

Vision Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
CLIP-ViT-L	59.44	24.84	6.46	0.584	0.645
+iREPA	69.70	21.26	6.64	0.601	0.644
DINOv2-B	64.92	22.75	6.54	0.593	0.653
+iREPA	70.88	21.40	6.77	0.597	0.653
DINOv2-L	66.65	22.46	6.53	0.594	0.650
+iREPA	72.41	20.99	6.84	0.600	0.650
DINOv3-B16	62.57	23.91	6.77	0.584	0.647
+iREPA	75.48	19.65	6.68	0.606	0.649
DINOv3-H16+	61.91	24.82	6.57	0.579	0.645
+iREPA	71.76	21.80	6.84	0.592	0.655
DINO-B	58.37	24.71	6.32	0.586	0.635
+iREPA	57.55	24.41	6.43	0.587	0.638
LangPE-G	58.42	25.37	6.32	0.583	0.641
+iREPA	66.88	21.62	6.65	0.604	0.649
MoCov3-B	57.61	25.60	6.25	0.579	0.640
+iREPA	61.41	22.80	6.46	0.596	0.639
PE-B	58.86	25.78	6.44	0.58	0.64
+iREPA	69.50	21.15	6.66	0.601	0.659
PE-G	56.09	26.84	6.45	0.568	0.641
+iREPA	70.43	21.52	6.58	0.598	0.651
SpatialPE-B	59.19	24.28	6.37	0.586	0.646
+iREPA	61.50	22.71	6.41	0.604	0.644
SpatialPE-L	62.13	23.87	6.51	0.59	0.65
+iREPA	68.83	21.07	6.59	0.607	0.651
WebSSL-1B	63.20	23.48	6.33	0.593	0.646
+iREPA	73.54	19.74	6.63	0.612	0.646
WebSSL-2B	66.56	22.27	6.69	0.596	0.654
+iREPA	72.13	20.47	6.89	0.609	0.648
WebSSL-300M	62.06	23.64	6.48	0.590	0.646
+iREPA	66.83	21.46	6.71	0.606	0.653
WebSSL-3B	65.91	22.55	6.53	0.598	0.654
+iREPA	71.98	20.64	6.71	0.607	0.650

Table 5: **SiT-B/2 performance across vision encoders at 400K iterations.** iREPA improves generation quality for every one of the vision encoders considered, with consistent FID and IS gains. All baselines are reported using vanilla-REPA (Yu et al., 2024) for training.

E.2 SiT-L/2 RESULTS

Vision Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
CLIP-ViT-L	107.7	10.6	5.20	0.682	0.643
+iREPA	115.8	9.41	5.15	0.689	0.653
DINOv2-B	113.3	9.61	5.11	0.684	0.653
+iREPA	113.9	9.36	5.11	0.690	0.655
DINOv3-B16	115.3	9.41	5.24	0.687	0.651
+iREPA	118.1	9.06	5.25	0.685	0.650
DINOv3-H16+	110.7	10.6	5.32	0.677	0.652
+iREPA	120.5	9.64	5.29	0.681	0.663
LangPE-G	102.8	11.2	5.13	0.684	0.637
+iREPA	112.7	9.37	5.02	0.690	0.653
MoCov3-B	102.1	11.0	5.14	0.685	0.635
+iREPA	104.7	10.1	5.00	0.690	0.641
PE-G	98.38	12.6	5.31	0.670	0.646
+iREPA	117.1	9.65	5.33	0.683	0.651
WebSSL-1B	107.9	10.5	5.13	0.686	0.644
+iREPA	121.5	8.62	5.01	0.696	0.653

Table 6: **SiT-L/2 performance across vision encoders at 400K iterations.** iREPA improves generation quality for every encoder, with especially pronounced gains in FID and IS, mirroring the SiT-B/2 results and confirming that the improvement holds at this intermediate model scale. All baselines are reported using vanilla-REPA (Yu et al., 2024) for training.

F ADDITIONAL EXPERIMENTAL RESULTS

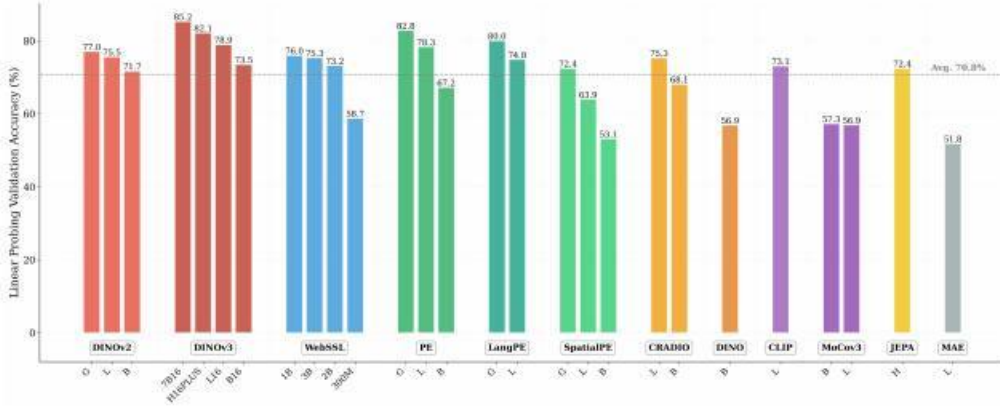


Figure 14: **Global information in mean patch tokens.** Beyond the [CLS] token, the mean of the patch tokens itself carries substantial global semantic information. While this raises global performance, it also erodes the contrast between individual patch tokens and thereby impedes spatial structure transfer. Removing part of this global component (the mean of the patch tokens) is precisely what strengthens spatial structure transfer during representation alignment (§4).

Vision Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
DINOv2-B	262.97	1.98	4.60	0.799	0.610
+iREPA (Ours)	268.79	1.93	4.59	0.799	0.600
DINOv3-B	261.18	1.99	4.58	0.799	0.609
+iREPA (Ours)	272.41	1.89	4.58	0.799	0.600
WebSSL-1B	250.53	2.24	4.61	0.809	0.580
+iREPA (Ours)	271.59	1.90	4.58	0.798	0.609
PE-G	238.37	2.44	4.57	0.805	0.585
+iREPA (Ours)	275.36	1.93	4.59	0.796	0.606

Table 7: **Generation quality with classifier-free guidance.** REPA vs. iREPA with CFG (scale 2.0) across vision encoders. All experiments use SiT-XL/2 trained for 400K iterations with 250 sampling steps. iREPA improves both IS and FID for every encoder, confirming that its gains persist under the guidance regime used at inference.

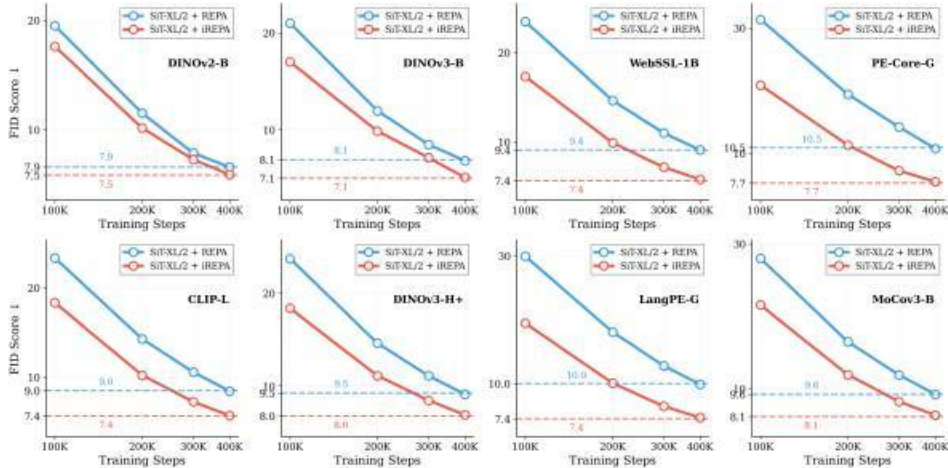


Figure 15: **Accentuating spatial features consistently improves convergence speed.** FID convergence results for SiT-XL/2 under REPA and iREPA; iREPA converges faster throughout training.

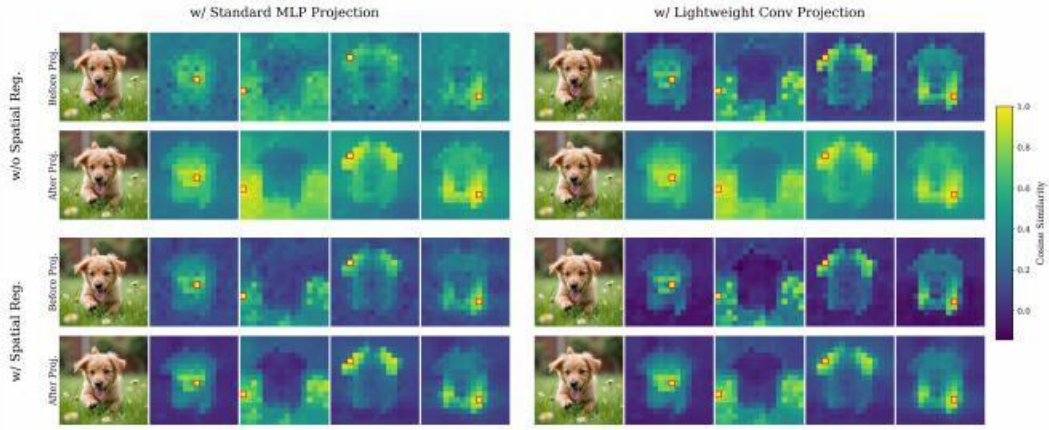


Figure 16: **Visualizing the impact of two straightforward improvements that enhance spatial feature transfer.** **First**, the standard MLP projection layer (top-left) loses spatial information when transferring features from the pretrained encoder (after projection) to the diffusion model (before projection). A simpler convolutional layer preserves this spatial information far more faithfully (top-right). **Second**, vision encoder features often exhibit limited spatial contrast (§2), so that tokens in one semantic region (*e.g.*, dog) attain substantial cosine similarity with unrelated tokens (*e.g.*, background). We address this with a simple spatial regularization layer that accentuates the spatial contrast in the representation, thereby improving generation performance. The best results are obtained by combining the two (see Table 2).

G IMPLEMENTATION DETAILS

Training setup. Unless otherwise specified, we follow the setup of REPA (Yu et al., 2024). All training is conducted on the ImageNet (Deng et al., 2009) training split. For preprocessing, we adopt the ADM (Dhariwal & Nichol, 2021) protocol, center-cropping and resizing each image to 256×256 resolution, and we use the `stabilityai/sd-vae-ft-mse` VAE (AI, n.d.) throughout diffusion training and inference. The spatial normalization layer uses $\gamma \in [0.6, 0.8]$, and the projection layer is a convolutional layer with kernel size 3 and padding 1. Optimization uses AdamW (Kingma & Ba, 2014; Loshchilov, 2017) with a constant learning rate of 1×10^{-4} and a global batch size of 256. For efficiency and stability, we train with `bfloat16` mixed precision and `torch.compile`, and apply gradient clipping together with an exponential moving average (EMA) of the generative model. Aside from the two iREPA modifications, all other hyperparameters are held fixed at their REPA defaults, so that the reported gains are attributable to the spatial modifications alone.

For the REPA-E (Leng et al., 2025a) and JiT (Li & He, 2025a) experiments, we build directly on the official open-source implementations. For the MeanFlow (Geng et al., 2025) experiments, we obtained the implementation (including REPA) from the original authors and adapted it by incorporating the two minimal iREPA modifications (§4), leaving the remainder of the recipe unchanged.

Evaluation. For image generation, we follow the ADM evaluation protocol (Dhariwal & Nichol, 2021) exactly. We report generation quality using Fréchet inception distance (gFID) (Heusel et al., 2017), structural FID (sFID) (Nash et al., 2021), inception score (IS) (Salimans et al., 2016), and precision (Prec.) and recall (Rec.) (Kynkäänniemi et al., 2019), each measured over 50K generated images. For sampling, we follow REPA (Yu et al., 2024) and use the SDE Euler–Maruyama sampler with 250 steps; for JiT (Li & He, 2025a) we use 50 inference steps, matching its official implementation.

H MORE DISCUSSION ON RELATED WORK

Classical spatial features in computer vision. Spatial feature extraction has long been foundational to computer vision, with classical descriptors such as SIFT (Lowe, 1999), HOG (Dalal & Triggs, 2005), SURF (Bay et al., 2006), and ORB (Rublee et al., 2011) furnishing robust local representations for tasks ranging from object detection to image matching. Although these handcrafted features excel at capturing local spatial patterns and geometric invariances, recent generative modeling has concentrated almost exclusively on modern self-supervised representations selected for strong global classification performance, such as DINOv2 (Oquab et al., 2024) and CLIP (Radford et al., 2021). Our findings motivate a different perspective: because spatial structure preservation is what governs generation quality, even classical spatial descriptors improve diffusion training when properly aligned. This reinstates the full spectrum of spatial feature extractors—from traditional handcrafted descriptors to modern learned representations—as viable alignment targets, provided they maintain strong spatial coherence.

Pretrained visual encoders for generative models. Pretrained visual encoders have served generative models in several capacities: as discriminators that accelerate GAN convergence (Goodfellow et al., 2020; Sauer et al., 2021; 2022; 2023; Radford et al., 2021), as teachers in adversarial distillation for diffusion models (Sauer et al., 2024; Kang et al., 2024), and, most recently, as alignment targets. In GANs (Goodfellow et al., 2020), pretrained features have both improved convergence speed and enabled scaling to large datasets, as demonstrated by StyleGAN-XL (Sauer et al., 2022) and StyleGAN-T (Sauer et al., 2023) using CLIP (Radford et al., 2021) features. For diffusion models, adversarial distillation leverages pretrained encoders to guide student networks toward higher-fidelity samples, yielding clear perceptual gains. Most directly, REPA (Yu et al., 2024) aligns diffusion features with an external encoder, establishing that representation alignment improves both convergence and generation quality. We build on this direction but relocate the emphasis from discriminative capability to spatial structure preservation—the property our analysis identifies as decisive.

Denosing transformers. Transformer architectures have become the dominant backbone for scalable generative modeling, spanning diffusion transformers (Peebles & Xie, 2023) and flow-matching variants (Ma et al., 2024). Recent architectures push this backbone in several directions: GenTron (Chen et al., 2023) scales transformers beyond 3B parameters for text-to-image synthesis, U-ViT (Bao et al., 2023) shows that pure transformer backbones without U-Net inductive biases remain competitive,

and ARDiT (Tang et al., 2024) and DART (Jing et al., 2024) explore autoregressive formulations that couple denoising with sequential generation to trade off quality against speed. Across these advances, training such models from scratch remains computationally expensive, often demanding millions of iterations to reach good generation quality. Representation alignment methods such as REPA (Yu et al., 2024) have shown that pretrained features can dramatically accelerate this convergence, yet the mechanism behind the acceleration remained unclear. Our analysis resolves it: the benefit stems from preserving spatial structure rather than from semantic alignment—an account that explains why certain encoders accelerate training more than others and that directly informs the design of more effective alignment strategies. While (Wang et al., 2025c) propose improvements to REPA training through early stopping, they hypothesize that “REPA predominantly distills global semantic information while leaving structural information untapped.” In contrast, our analysis reveals that spatial structure (not global semantic information) already plays a very significant role in the effectiveness of REPA (§2, §3). Thus, while we indeed find that spatial structure remains underexploited (§4) in REPA, we surprisingly find that majority of improvements in REPA are already coming from introducing an inductive bias for spatial structure (not global information).

Denoising as a self-supervised learning task. The connection between denoising and representation learning has been examined from several angles. Early work by Abstreiter et al. (2021) extended the diffusion objective to representation learning, showing that denoising naturally acquires meaningful features; SODA (Hudson et al., 2024) couples a diffusion model with an information bottleneck to learn compact representations; and Chen et al. (2024) deconstruct diffusion models to show that a simple denoising autoencoder already attains strong self-supervised performance. Wang & He (2025) introduce a dispersive loss that encourages internal representation diversity and improves generation without external encoders, and Jiang et al. (2025) propose Self-Representation Alignment (SRA), which aligns a diffusion transformer’s latent features across noise levels to supply self-guidance without an auxiliary model. Collectively, these works establish denoising as a fundamental self-supervised task that naturally encourages robust features. Our findings sharpen this view: aligning diffusion models with the spatial representations of self-supervised encoders improves both generative and discriminative capability, and pinpoints spatial structure preservation as the factor underlying the synergy.

Scaling self-supervised vision encoders. Recent years have brought remarkable progress in scaling self-supervised vision models to unprecedented sizes and datasets. DINOv3 (Siméoni et al., 2025) trains a 7B-parameter ViT on 1.7 billion unlabeled images by aligning representations across augmentations; WebSSL (Fan et al., 2025) shows that models trained on more than 2 billion images match language-supervised models such as CLIP (Radford et al., 2021) on vision-language tasks without any language supervision; and C-RADIO (Heinrich et al., 2024) distills multiple teachers into versatile encoders that excel across visual domains. I-JEPA (Assran et al., 2023) learns by predicting masked regions in representation space rather than reconstructing pixels, SAM (Ravi et al., 2024) specializes in promptable segmentation via large-scale supervision, and the Perception Encoder family (Bolya et al., 2025; Cho et al., 2025) shows that intermediate features often surpass final representations for dense prediction. These models are almost invariably evaluated on global tasks such as image classification, yet our work exposes a crucial gap: strong discriminative performance does not translate into better generation. It is instead the encoders that preserve spatial structure—irrespective of their classification accuracy—that provide the greatest benefit for diffusion training. This calls for the evaluation criteria of self-supervised encoders to be reconsidered for generative applications, treating spatial coherence as no less important than semantic understanding.

I NOTE ON LLM USAGE

We use GPT-5 (OpenAI, 2025) to explore variations of the spatial structure metrics discussed in §3. All figures in the paper are generated directly from our experiment logs and checkpoints using Claude-Code (Anthropic, 2025). We additionally rely on LLM assistance for surveying and formulating the related work in Appendix H, and use Cursor in places to assist with paper writing.

J ADDITIONAL DISCUSSION ON SPATIAL STRUCTURE METRICS

Spatial structure metrics can measure the representation gap. Yu et al. (2024) measure the representation gap through linear probing, invoking the resulting increase in validation accuracy to explain the effectiveness of REPA. We show that spatial structure metrics (SSM) measure this gap equally well while additionally accounting for improvements across models: as Figure 17 demonstrates, representation alignment closes the gap between the SSM of a pretrained encoder such as DINOv2 and that of the diffusion features—recasting the representation gap in the spatial terms that our analysis shows to be decisive.

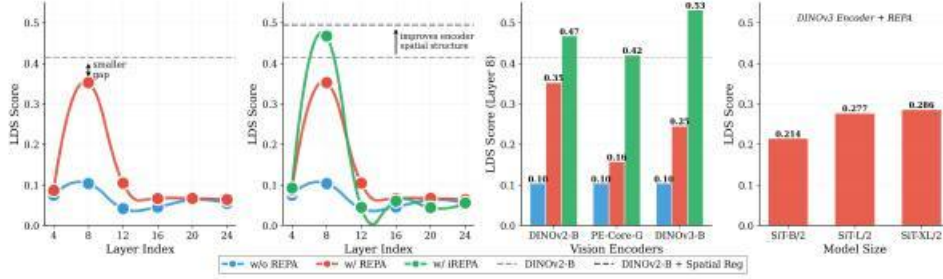


Figure 17: Representation alignment as bridging the spatial feature gap. **a)** Representation alignment can itself be read as bridging the spatial feature gap between the diffusion and vision-encoder patch features. **b)** iREPA (§4) accentuates the spatial features of the vision encoder—at the cost of some global information—thereby achieving better generation performance. **c)** iREPA improves performance consistently across vision encoders. **d)** The spatial-structure improvements scale with diffusion model size. All results are reported with SiT-XL/2 at 400k iterations.

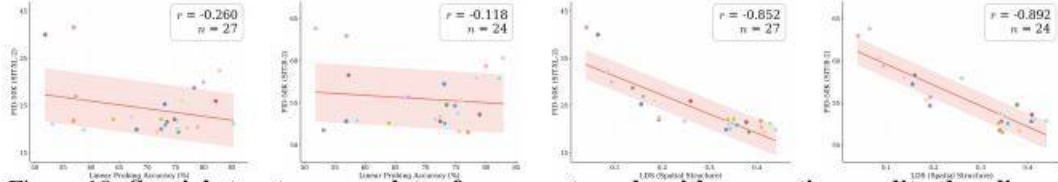


Figure 18: Spatial structure correlates far more strongly with generation quality than linear probing. Correlation analysis across 27 vision encoders. **Left two:** Linear probing accuracy vs. FID for SiT-XL-2 (Pearson $r = -0.26$) and SiT-B-2 ($r = -0.12$) is only weakly correlated. **Right two:** Spatial structure (LDS) vs. FID for SiT-XL-2 ($r = -0.85$) and SiT-B-2 ($r = -0.89$) is strongly correlated at both scales. See Figure 10 in the Appendix for detailed plots with encoder labels.

K ADDITIONAL ANECDOTAL COMPARISONS

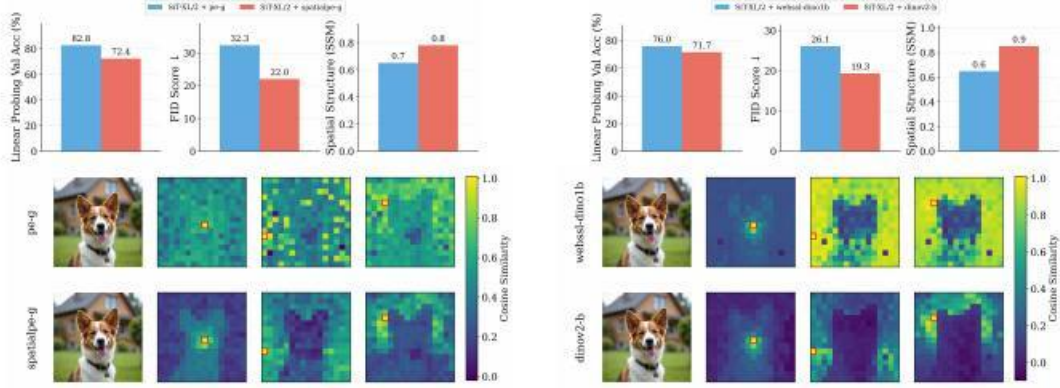


Figure 19: **Motivating anecdotes from recent SSL encoders — spatial structure matters.** **Top row:** Metrics comparison showing inverse relationship between ImageNet accuracy and generation quality. **Left:** PE-g achieves higher accuracy (82.8% vs 72.4%) but worse FID (32.3 vs 22.0) than Spatial-PE-g, with much lower spatial structure (LDS: 0.1 vs 0.4). **Right:** WebSSL-dino1b shows higher accuracy (76.0% vs 71.7%) but worse FID (26.1 vs 19.3) than DINOv2-b, with weaker spatial structure (LDS: 0.2 vs 0.4). **Bottom row:** Spatial cosine similarity visualizations confirm that encoders with better generation (Spatial-PE-g, DINOv2-b) maintain clear spatial coherence, while those optimized for classification (PE-g, WebSSL) lose spatial structure.

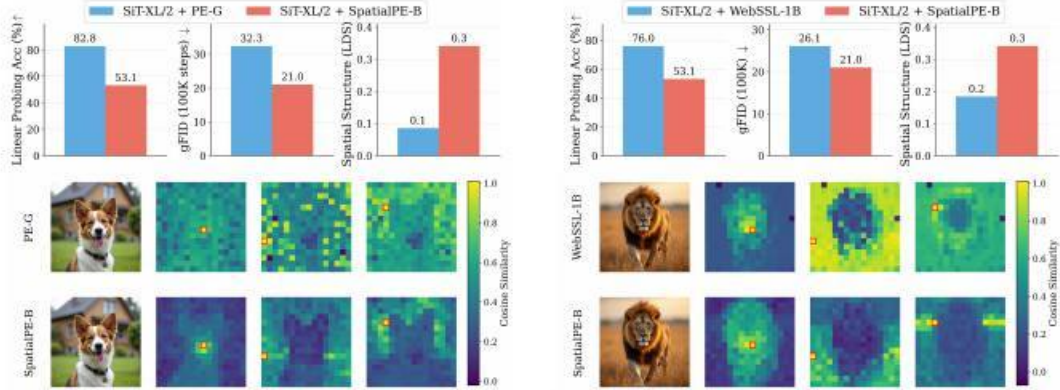


Figure 20: **Motivating examples — spatial structure matters.** **Top:** Metrics comparison showing inverse relationship between ImageNet accuracy and generation quality. **Left:** PE_{core}-G, despite having significantly higher validation accuracy (82.8% vs. 53.1%), shows worse generation quality compared to PE_{Spatial}-B (Bolya et al., 2025). **Right:** Similarly, WebSSL-1B (Fan et al., 2025) also shows much better global performance (76.0% vs. 53.1%), but worse generation. **Bottom:** We find that spatial structure instead provides a better predictor of generation quality than global performance. See §3 for spatial structure metric. All results reported at 100K using SiT-XL/2 and REPA.

L FURTHER DISCUSSION

Role of Spatial Normalization. We visualize the effect of spatial normalization on vision encoder representations. Pretrained encoders frequently carry a substantial global component that suppresses the contrast among spatial tokens, so that tokens in one semantic region become highly correlated with tokens in unrelated regions (e.g., background) and lose their spatial distinctiveness. Spatial normalization removes this global overlay and sharpens the contrast between spatial tokens; the representation thereby retains its local spatial structure while shedding the global information that otherwise interferes with generation quality.

The following figures illustrate this effect across several examples, showing how spatial normalization reshapes the feature representations to emphasize spatial structure:

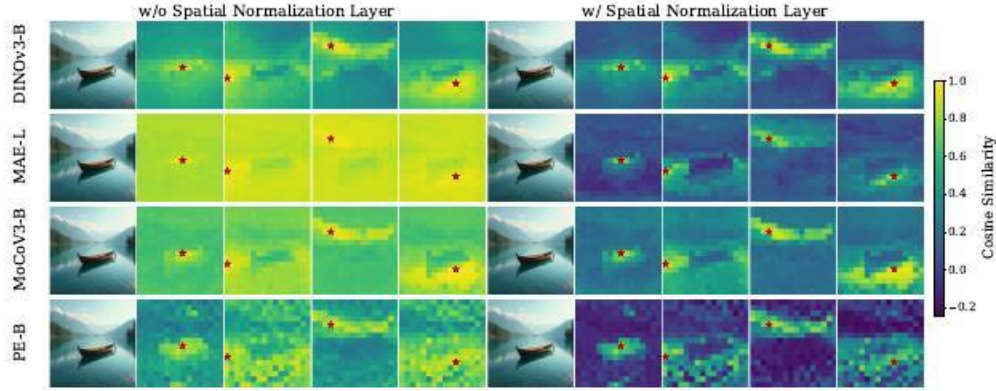


Figure 21: Visualizing impact of spatial normalization (Example 1). The heatmaps show token similarity patterns before and after spatial normalization. Without normalization (left), global components create high correlations across unrelated regions. With spatial normalization (right), local spatial structure is enhanced while reducing global interference, resulting in more distinct semantic boundaries.

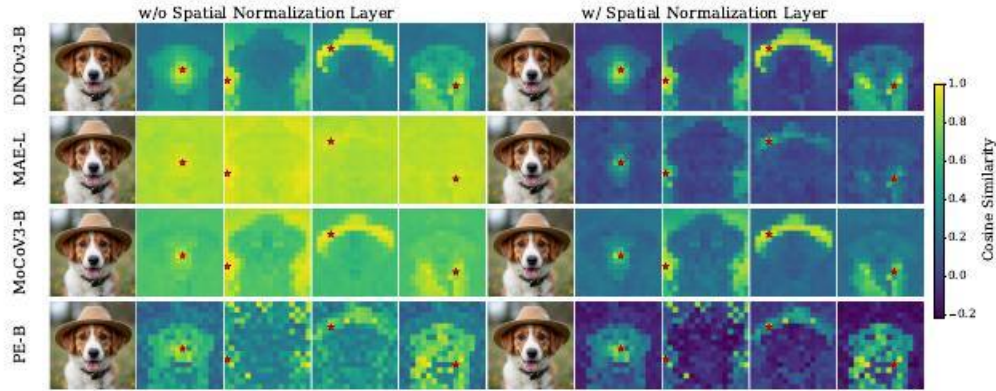


Figure 22: Visualizing impact of spatial normalization (Example 3). The heatmaps show token similarity patterns before and after spatial normalization. Without normalization (left), global components create high correlations across unrelated regions. With spatial normalization (right), local spatial structure is enhanced while reducing global interference, resulting in more distinct semantic boundaries.

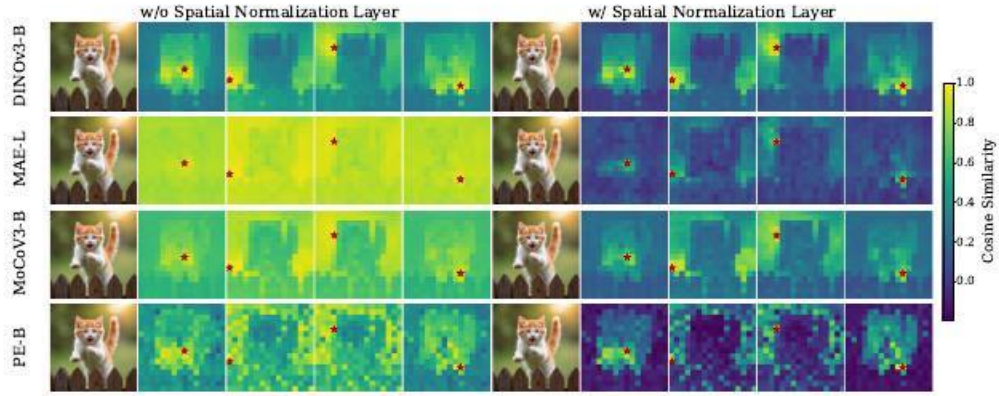


Figure 23: Visualizing impact of spatial normalization (Example 2). Feature similarity maps demonstrate how spatial normalization improves spatial contrast. The original features (left) exhibit a global overlay that reduces distinction between foreground and background regions. After normalization (right), spatial tokens become more locally coherent, with clearer separation between different semantic regions.

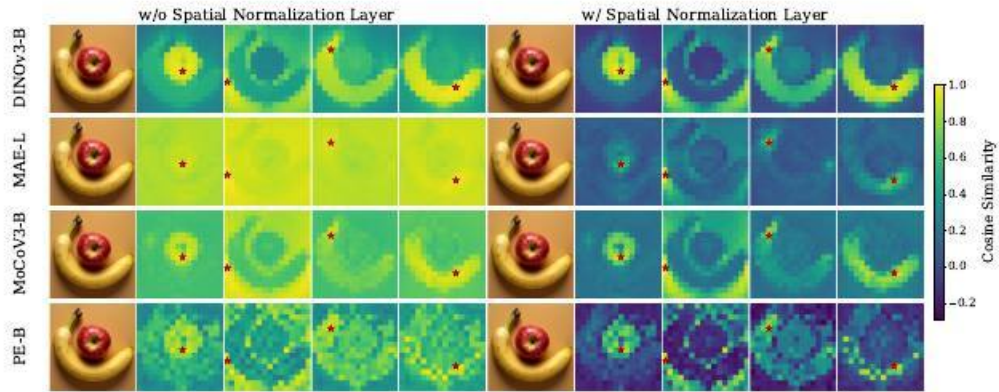


Figure 24: **Visualizing impact of spatial normalization** (Example 4). The heatmaps show token similarity patterns before and after spatial normalization. Without normalization (left), global components create high correlations across unrelated regions. With spatial normalization (right), local spatial structure is enhanced while reducing global interference, resulting in more distinct semantic boundaries.

M ADDITIONAL RESULTS

M.1 ADDITIONAL RESULTS AT HIGHER RESOLUTIONS

Setup. We conduct further experiments on ImageNet-512 (Deng et al., 2009) to assess how the proposed spatial improvements generalize to higher resolutions. Following the REPA (Yu et al., 2024) setup, we report results across a range of pretrained encoders (DINOv2, DINOv3, WebSSL, PE, etc.), using SiT-XL and SiT-B with 50 NFE at inference and without classifier-free guidance. Fig. 25 contrasts REPA before and after the spatial improvements (iREPA). The spatial improvements transfer cleanly to the higher resolution, delivering consistent gains across every pretrained encoder considered (DINOv2, DINOv3, WebSSL, PE, etc.) and confirming that the principle is not tied to the 256×256 setting.

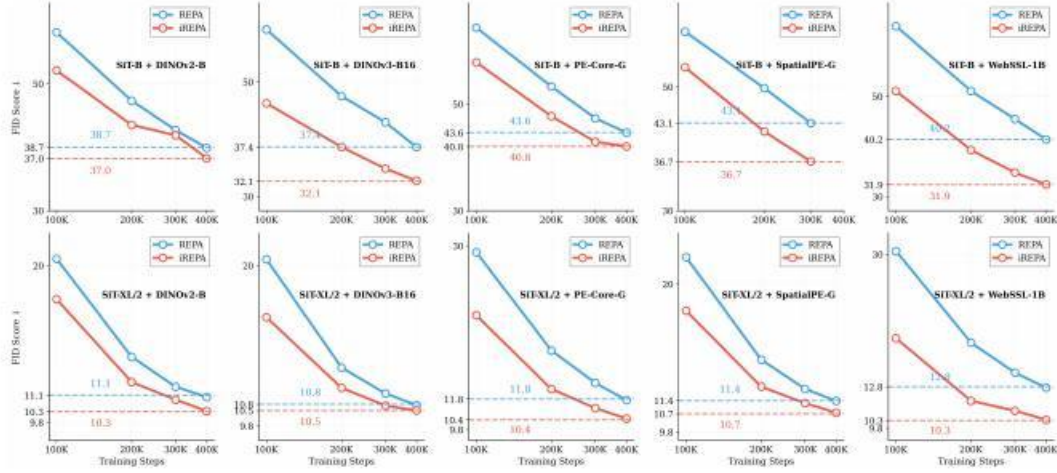


Figure 25: **Convergence results at 512x512 resolution.** Accentuating spatial features consistently improves convergence speed across resolutions, holding for both ImageNet-256 (Figure 7) and ImageNet-512 (above)—evidence that the benefit is resolution-invariant.

M.2 ADDITIONAL RESULTS ON MULTIMODAL T2I TASKS

Setup. To probe the generalizability of the proposed spatial improvements beyond ImageNet, we further conduct extensive experiments on multimodal text-to-image (T2I) tasks. Following REPA (Yu et al., 2024), we adopt MMDiT (Esser et al., 2024) as the diffusion backbone and apply REPA with a range of pretrained vision encoders (DINOv2, CLIP, WebSSL, PE, etc.). As in REPA (Yu et al., 2024), all models are trained for 150K steps with a batch size of 256 and evaluated with an ODE sampler at 50 NFE. Fig. 26 contrasts REPA before and after the spatial improvements (iREPA). The spatial improvements extend to the multimodal T2I setting as well, yielding consistent gains across every pretrained encoder considered (DINOv2, CLIP, WebSSL, PE, etc.)—demonstrating that the principle carries beyond class-conditional ImageNet generation to conditional multimodal synthesis.

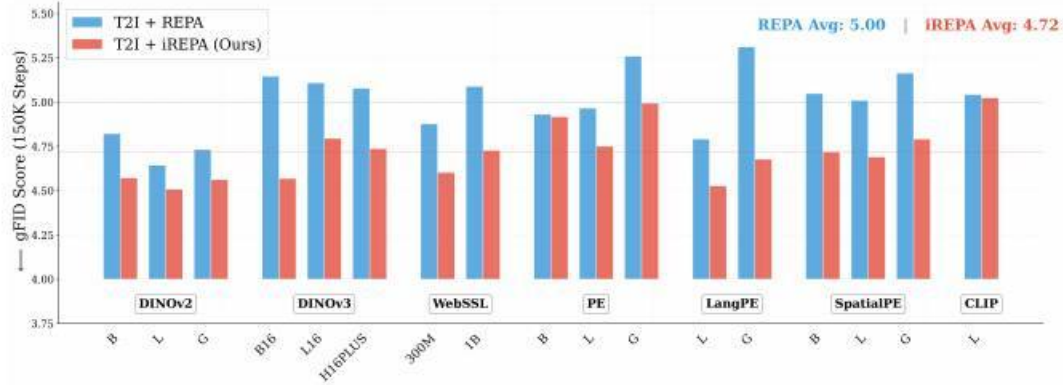


Figure 26: **Text-to-image generation across encoder variants.** Accentuating the transfer of spatial features from the target representation to the diffusion features consistently improves convergence speed for both ImageNet (see §4) and multimodal T2I tasks (above). The gains hold uniformly across the choice of pretrained encoder (DINOv2, CLIP, WebSSL, PE, etc.).

M.3 CORRELATION ANALYSIS WITHOUT OUTLIERS (MoCOv3-L AND MAE-L)

We repeat the correlation analysis from Section 3 after removing MoCOv3-L and MAE-L, which were identified as outliers. Figure 28 reports the updated correlations across model sizes. Spatial structure continues to correlate far more strongly with generation performance than linear probing accuracy. Notably, once the outliers are removed, linear probing exhibits a small *positive* correlation with gFID—that is, higher linear probing accuracy is associated with *worse* generation. This trend directly reinforces the observations of Sec. 2, in which target representations with higher global semantic performance (linear probing accuracy) frequently yield similar or worse generation under representation alignment (REPA).

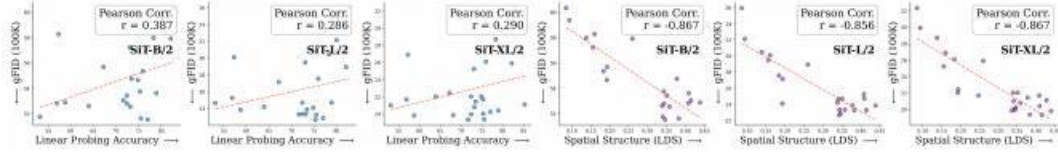


Figure 27: Correlation analysis without outliers. After removing the outliers (MoCOv3-L and MAE-L), spatial structure still correlates far more strongly (Pearson $|r| > 0.85$) with generation performance than linear probing accuracy (Pearson $|r| < 0.38$) across all model sizes (B, L, XL). Notably, once the outliers are removed, linear probing exhibits a small *positive* correlation with gFID—higher linear probing accuracy is associated with *worse* generation. This trend directly reinforces the observations of Sec. 2, in which target representations with higher global semantic performance (linear probing accuracy) frequently yield similar or worse generation under representation alignment (REPA).

M.4 ADDITIONAL QUALITATIVE RESULTS

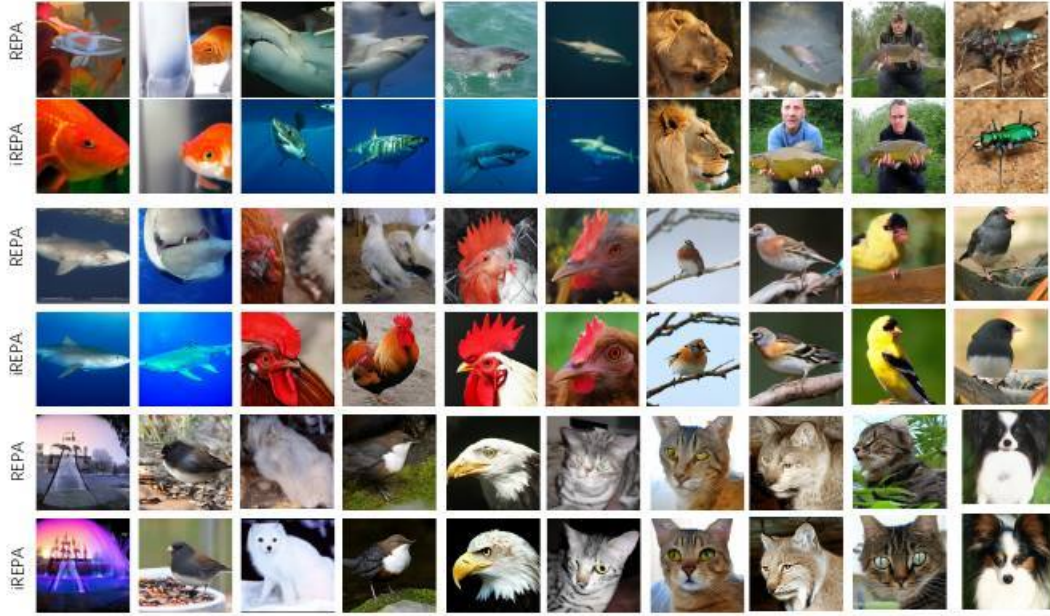


Figure 28: **Additional Qualitative Results** comparing generation outputs before and after the spatial improvements with REPA. All results use PE-G (Bolya et al., 2025) as the pretrained vision encoder, 400K steps (80 epochs), and a classifier-free guidance scale of 4.0. Mirroring the quantitative improvements (§4), the spatial improvements (iREPA) also enhance the visual quality and spatial coherence of the generated outputs.

M.5 ADDITIONAL RESULTS WITH ALTERNATIVE EVALUATION METRICS

Beyond the traditional evaluation metrics (Inception Score, FID, sFID, Precision, Recall), we further verify the robustness of the proposed findings using alternative measures of generation quality (Stein et al., 2023; Yang et al., 2023; Kynkäänniemi et al., 2023; Jayasumana et al., 2024). Specifically, we adopt the CMMD metric (Jayasumana et al., 2024) with the PyTorch implementation of (Paul, 2024), using the standard reference set from the OpenAI ADM evaluation suite (Dhariwal & Nichol, 2021) as reference images. We then re-verify the key findings of both §3 and §4 under CMMD.

Spatial structure correlates far more strongly with generation performance than linear probing.

To assess the robustness of the analysis in §3, we repeat the large-scale correlation analysis across vision encoders using the CMMD metric (Jayasumana et al., 2024). Results are reported in Figure 29, using SiT-B/2 (100K steps) and REPA. Every spatial metric correlates far more strongly (Pearson $|r| > 0.88$) with generation performance (CMMD) than linear probing does (Pearson $|r| = 0.074$), establishing that the central findings of §3 hold irrespective of whether generation quality is measured by FID or CMMD.

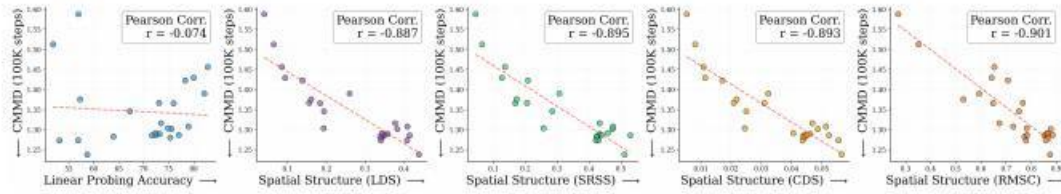


Figure 29: **Correlation analysis with the CMMD metric.** We repeat the correlation analysis of §3 with the CMMD metric (Jayasumana et al., 2024) in place of gFID. All results use SiT-B/2 (100K steps) and REPA. Every spatial metric correlates far more strongly (Pearson $|r| > 0.88$) with generation performance (CMMD) than linear probing does (Pearson $|r| = 0.074$), confirming that the key findings of §3 are robust to the choice of evaluation metric.

Accentuating the transfer of spatial features improves generation quality. We next examine the generalizability of the analysis in §4 under the CMMD (Jayasumana et al., 2024) metric, alongside the traditional evaluation metrics (gFID, sFID, IS, etc.). Results are reported in Table 8. Across the range of vision encoders, accentuating the transfer of spatial features (iREPA) improves convergence speed under both CMMD and the traditional metrics (IS, FID, sFID, Prec., Recall), and the improvement holds consistently both with and without classifier-free guidance (CFG).

Vision Encoder	Steps	w/o CFG						w/ CFG					
		CMMD↓	IS↑	FID↓	sFID↓	Prec.↑	Rec.↑	CMMD↓	IS↑	FID↓	sFID↓	Prec.↑	Rec.↑
DINOv2-B	100K	0.702	69.20	19.3	5.89	0.64	0.61	0.529	157.2	6.35	5.91	0.77	0.54
+iREPA	100K	0.652	77.92	16.9	6.26	0.66	0.61	0.484	179.3	5.15	6.23	0.78	0.54
DINOv2-B	400K	0.455	127.4	7.76	5.06	0.70	0.66	0.320	263.0	1.98	4.60	0.80	0.61
+iREPA	400K	0.438	128.6	7.52	4.89	0.71	0.65	0.310	268.8	1.93	4.59	0.80	0.60
DINOv3-B	100K	0.749	63.64	21.4	6.14	0.63	0.60	0.571	144.0	7.57	6.09	0.76	0.53
+iREPA	100K	0.651	78.79	16.2	6.14	0.66	0.61	0.481	181.9	4.87	6.10	0.78	0.55
DINOv3-B	400K	0.474	126.7	8.10	5.06	0.70	0.66	0.336	261.2	1.99	4.58	0.80	0.61
+iREPA	400K	0.441	132.9	7.13	4.93	0.71	0.66	0.314	272.4	1.89	4.58	0.80	0.60
WebSSL-1B	100K	0.825	53.87	25.5	6.57	0.61	0.59	0.622	124.0	9.59	6.37	0.76	0.52
+iREPA	100K	0.653	77.47	16.6	6.18	0.66	0.61	0.484	177.6	5.09	6.11	0.79	0.54
WebSSL-1B	400K	0.512	116.9	9.39	5.14	0.70	0.64	0.354	250.5	2.24	4.61	0.81	0.58
+iREPA	400K	0.445	130.8	7.48	4.91	0.70	0.65	0.311	271.6	1.90	4.58	0.80	0.61
PE-Core-G	100K	0.922	42.74	32.3	6.70	0.57	0.59	0.714	97.2	14.1	6.56	0.71	0.53
+iREPA	100K	0.697	75.01	18.1	6.03	0.64	0.61	0.525	176.8	5.66	6.08	0.77	0.54
PE-Core-G	400K	0.525	109.4	10.4	5.00	0.69	0.64	0.366	238.4	2.44	4.57	0.81	0.59
+iREPA	400K	0.458	132.0	7.78	5.02	0.70	0.65	0.322	275.4	1.93	4.59	0.80	0.61
CLIP-L	100K	0.790	54.07	25.2	6.65	0.61	0.60	0.605	124.5	9.77	6.59	0.75	0.52
+iREPA	100K	0.657	74.46	17.8	6.33	0.65	0.61	0.489	172.0	5.67	6.33	0.78	0.54
CLIP-L	400K	0.487	117.8	8.97	4.98	0.70	0.65	0.339	258.7	2.15	4.64	0.81	0.59
+iREPA	400K	0.442	130.1	7.42	5.03	0.71	0.65	0.307	271.8	1.96	4.64	0.80	0.61

Table 8: **Additional results with alternative evaluation metrics.** We report additional results under the CMMD metric (Jayasumana et al., 2024). All results use SiT-XL/2, batch size 256, and traditional REPA as the baseline; CMMD is computed with the PyTorch implementation of (Paul, 2024), using the standard reference set from the OpenAI ADM evaluation suite (Dhariwal & Nichol, 2021) as reference images. Across every vision encoder, accentuating the transfer of spatial features (iREPA) improves convergence speed under both CMMD and the traditional metrics (IS, FID, etc.), at both 100K and 400K steps and with and without CFG. This establishes that the key findings of §4 are robust to the choice of evaluation metric.

M.6 SPATIAL IMPROVEMENTS WITH SAM2

Table 9 and Figure 30 study the impact of spatial normalization (§4) on SAM2-S (46M) encoder features². As Table 9 shows, spatial normalization does improve SAM2 performance, though the margin is modest. The reason is instructive and consistent with our account: spatial normalization operates by removing the global component (the mean of the patch tokens) to enhance spatial contrast, and because SAM2 already carries little to no global information (validation accuracy < 24%), there is little global component left to remove, so the spatial contrast is only slightly sharpened (see Figure 30).

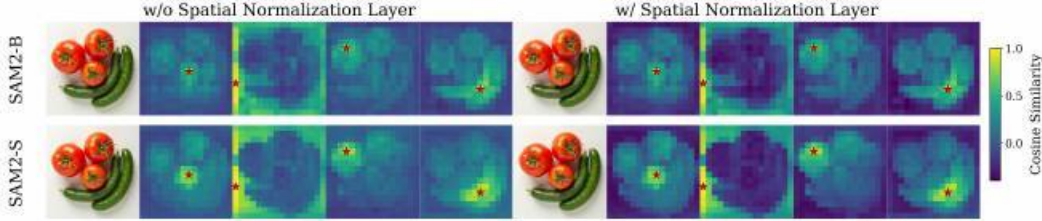


Figure 30: **Impact of spatial normalization on SAM2.** For SAM2, the spatial normalization layer does enhance spatial contrast, but only by a modest margin. The reason is consistent with our account: spatial normalization (§4) operates by removing the global component (the mean of the patch tokens), and because SAM2 already carries little to no global information (validation accuracy < 24%), there is little to remove, so the spatial contrast is only slightly sharpened.

Vision Encoder	Steps	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
SAM2-S	100K	50.69	25.32	6.52	0.631	0.588
+iREPA	100K	53.14	24.52	6.28	0.629	0.591
SAM2-S	400K	110.77	9.54	4.93	0.699	0.638
+iREPA	400K	114.62	9.10	4.89	0.704	0.640

Table 9: **Impact of spatial improvements on SAM2.** iREPA improves generation over baseline REPA at both 100K and 400K steps, though the margin is modest because SAM2 already carries little global information for spatial normalization to remove. All results use SiT-XL/2, without classifier-free guidance, SAM2-S (46M) as the vision encoder, and traditional REPA as the baseline.

²Note that similar to (Bolya et al., 2025), we use the intermediate output of vision encoder for SAM2 features and not the mask logits. As shown in (Bolya et al., 2025), while mask logits lead to sharper spatial maps, the mask logits itself are not suitable as a target representation.

M.7 ADDITIONAL RESULTS WITH FULL-FINETUNING ACCURACY INSTEAD OF LINEAR PROBING ACCURACY

Correlation analysis with full-finetuning accuracy instead of linear probing accuracy. We repeat the correlation analysis of §3 using validation accuracy after full-finetuning³ in place of linear probing accuracy. Results are reported in Fig. 31. All results use SiT-XL/2 (100K steps) and REPA; for full-finetuning validation accuracy, we finetune the entire vision encoder with a learning rate of $5e-5$, a warmup ratio of 0.1, and 30 epochs in total. Every spatial metric correlates far more strongly with generation performance than full-finetuning accuracy does. Strikingly, gFID exhibits a weak *positive* correlation with full-finetuning validation accuracy (Pearson $r = 0.317$)—as validation accuracy rises, gFID rises and generation deteriorates. This reinforces the observations of §2, in which target representations with higher global semantic performance frequently yield similar or worse generation under representation alignment (REPA), and confirms that the conclusion is unchanged whether global information is measured by linear probing or by full finetuning.

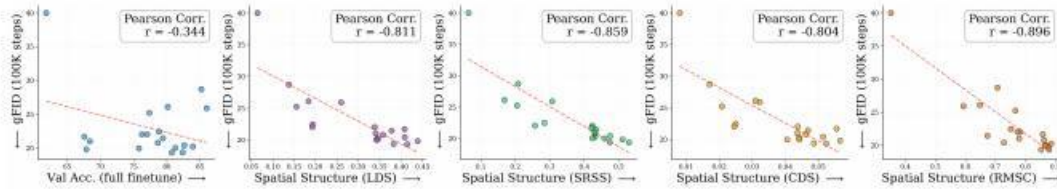


Figure 31: Correlation analysis with full-finetuning accuracy instead of linear probing accuracy. We repeat the correlation analysis of §3 using validation accuracy after full-finetuning in place of linear probing accuracy. All results use SiT-XL/2 (100K steps) and REPA. Every spatial metric correlates far more strongly with generation performance than full-finetuning accuracy does. Strikingly, gFID exhibits a weak positive correlation with full-finetuning validation accuracy (Pearson $r = 0.317$)—as validation accuracy rises, gFID rises and generation deteriorates. This reinforces the observations of §2, in which target representations with higher global semantic performance frequently yield similar or worse generation under representation alignment (REPA).

³Please note that while the final findings remain similar, in context of REPA linear probing might be more accurate for estimating global information in “pretrained” vision encoders. This is because REPA uses the “pretrained” encoder features themselves for regularization. Finetuning the encoder itself can impact the amount of global information. Same as REPA (Yu et al., 2024), we therefore use linear probing accuracy as default for measuring global information.

M.8 ADDITIONAL RESULTS ON PIXEL-SPACE DIFFUSION MODELS

Method	#Params	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
PixelFlow (Chen et al., 2025)	459M	24.67	54.33	9.71	-	0.58
PixDDT (Wang et al., 2025b)	434M	36.24	46.37	17.14	-	0.63
PixNerd (Wang et al., 2025a)	458M	43.01	37.49	10.65	-	0.62
DeCo w/o $\mathcal{L}_{\text{FreqFM}}$	426M	46.44	34.12	10.41	-	0.64
DeCo + REPA (Ma et al., 2025b)	426M	48.35	31.35	9.34	-	0.65
JiT (Li & He, 2025a)	459M	29.37	49.06	11.21	0.40	0.62
JiT+iREPA (Ours)	459M	50.72	29.19	9.42	0.51	0.65

Table 10: **Generation performance of pixel-space diffusion models.** Despite its simplicity, iREPA combined with JiT (Li & He, 2025a) outperforms recently proposed state-of-the-art pixel-space diffusion methods, including DeCo (Ma et al., 2025b). All results are reported at 200K training iterations with a batch size of 256, evaluated using 50-step Euler sampling without classifier-free guidance; results for DeCo are taken directly from (Ma et al., 2025b).